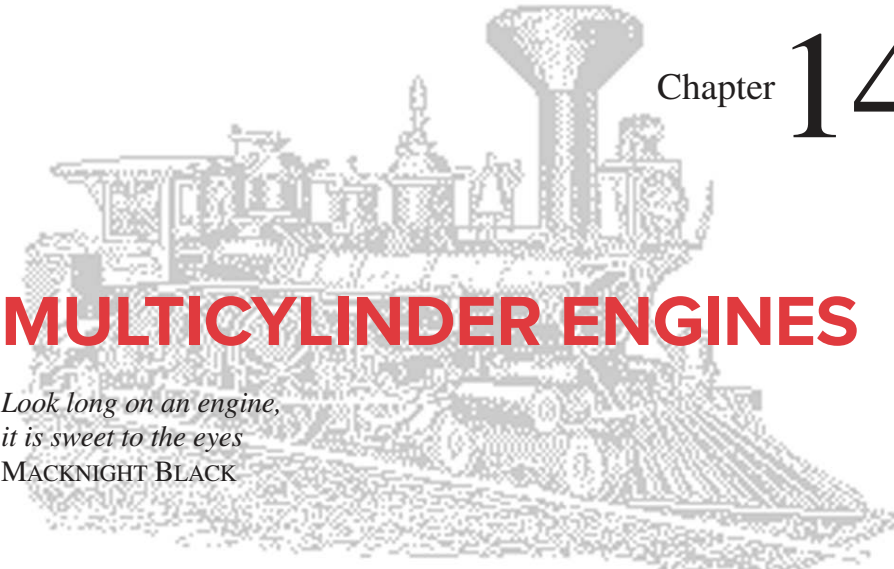




Chapter 14

MULTICYLINDER ENGINES

*Look long on an engine,
it is sweet to the eyes*

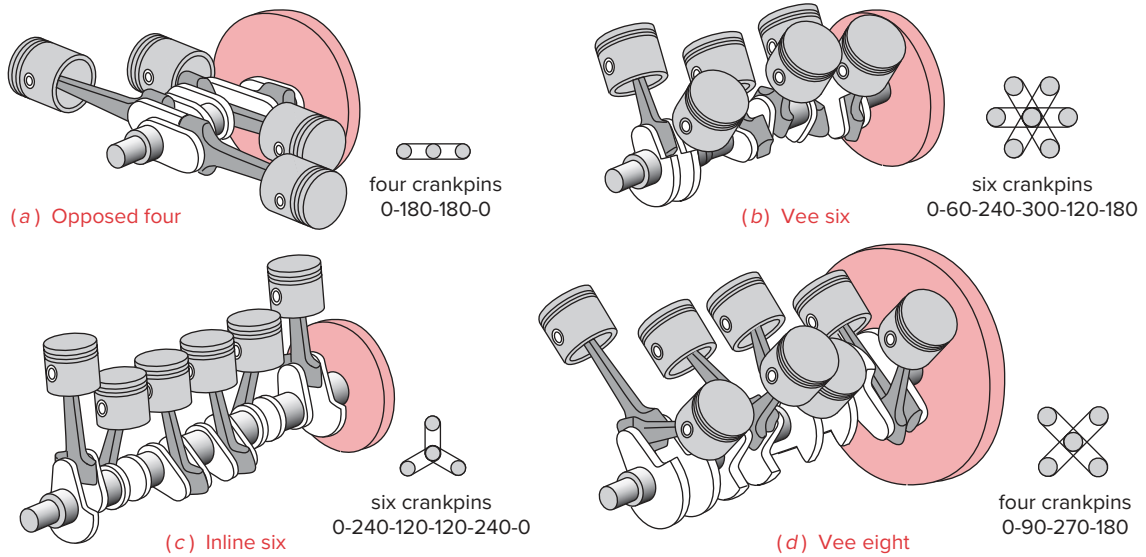
MACKNIGHT BLACK

14.0 INTRODUCTION *Watch a Video on Multicylinder Engines (44:25)**

The previous chapter discussed the design of the slider-crank mechanism as used in the single-cylinder internal combustion engine and piston pumps. We will now extend the design to multicylinder configurations. Some of the problems with shaking forces and torques can be alleviated by proper combination of multiple slider-crank linkages on a common crankshaft. Program LINKAGES, included with this text, will calculate the equations derived in this chapter and allow the student to exercise many variations of an engine design in a short time. Some examples are provided as disk files to be read into the program. These are noted in the text. The student is encouraged to investigate these examples with program LINKAGES in order to develop an understanding of and insight to the subtleties of this topic. A user manual for program LINKAGES is provided in the program and context-sensitive help is available within the program. See Appendix A for more information on using the programs that come with the book.

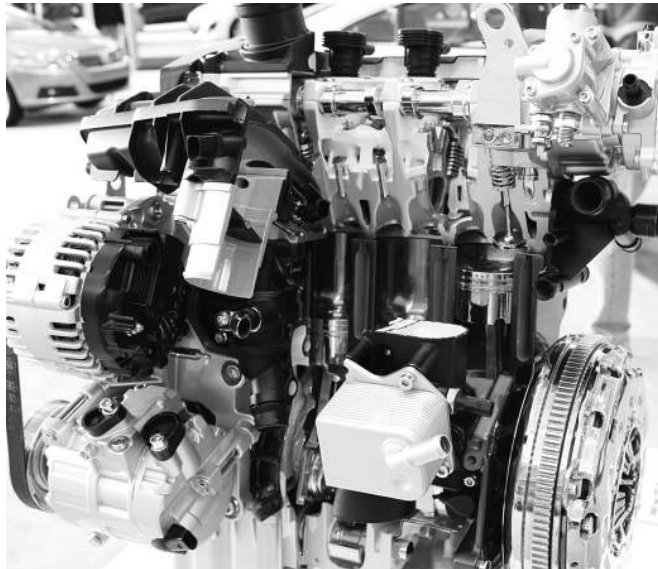
As with the single-cylinder case, we will not address the thermodynamic aspects of the internal combustion engine beyond the definition of the combustion forces necessary to drive the device presented in the previous chapter. We will concentrate on the engine's kinematic and mechanical dynamics aspects. It is not our intention to make an "engine designer" of the reader so much as to apply dynamic principles to a realistic design problem of general interest and also to convey the complexity and fascination involved in the design of a more complicated dynamic device than the single-cylinder engine.

* http://www.designofmachinery.com/DOM/Multicylinder_Engines.mp4

**FIGURE 14-1**

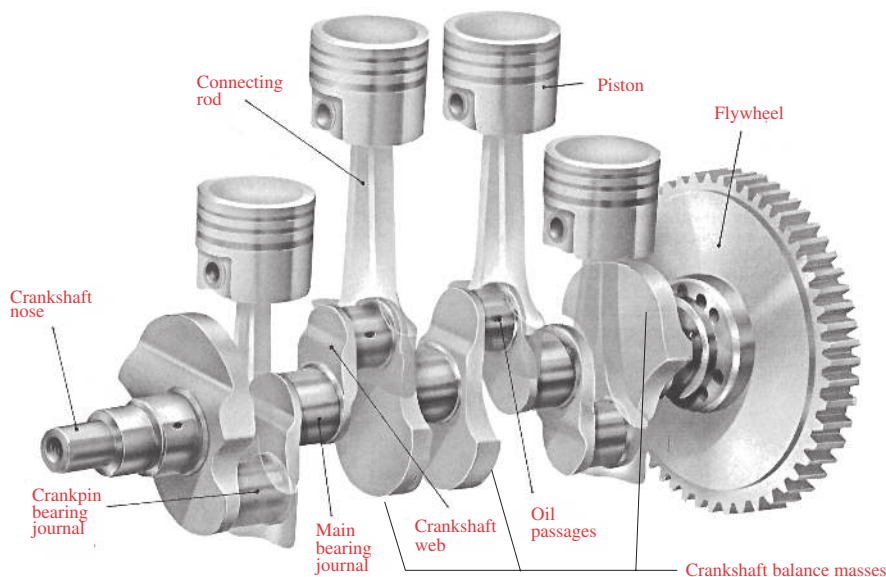
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Various multicylinder engine configurations

**FIGURE 14-2**

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Cutaway view of a four-stroke, VW-Audi four-cylinder inline engine

**FIGURE 14-3**

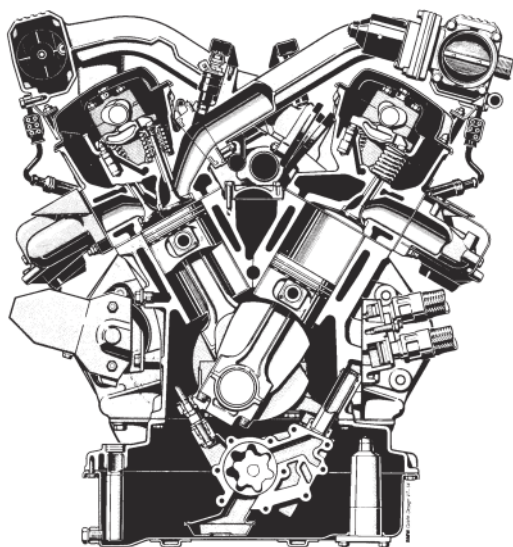
Crankshaft from an inline four-cylinder engine with pistons, connecting rods, and flywheel. Illustration copyright Eaglemoss Publications/Car Care Magazine. Reprinted with permission.

14.1 MULTICYLINDER ENGINE DESIGNS

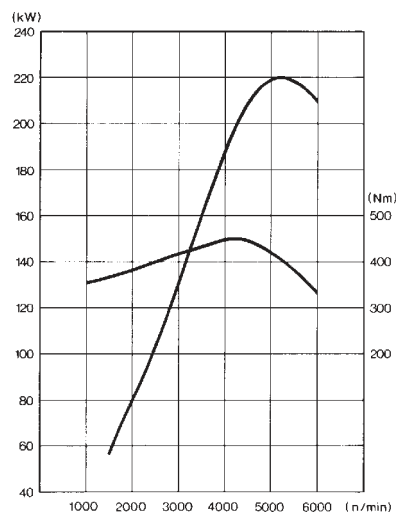
Multicylinder engines are designed in a wide variety of configurations from the simple inline arrangement to vee, opposed, and radial arrangements some of which are shown in Figure 14-1. These arrangements may use any of the stroke cycles discussed in Chapter 13: Clerk, Otto, or diesel.

INLINE ENGINES The most common and simplest arrangement is an inline engine with its cylinders all in a common plane as shown in Figure 14-2. Two-,* three-,* four-, five-, and six-cylinder **inline engines** are in common use. Each cylinder will have its individual slider-crank mechanism consisting of a crank, conrod, and piston. The cranks are formed together in a common **crankshaft** as shown in Figure 14-3. Each cylinder's crank on the crankshaft is referred to as a **crank throw**. These crank throws will be arranged with some **phase angle** relationship one to the other, in order to stagger the motions of the pistons in time. It should be apparent from the discussion of shaking forces and balancing in the previous chapter that we would like to have pistons moving in opposite directions to one another at the same time in order to cancel the reciprocating inertial forces. The optimum phase angle relationships between the crank throws will differ depending on the number of cylinders and the stroke cycle of the engine. There will usually be one or a small number of viable crank throw arrangements for a given engine configuration to accomplish this goal. The engine in Figure 14-2 is a four-stroke cycle, four-cylinder, VW-Audi inline engine with overhead camshaft. Figure 14-3 shows the crankshaft, connecting rods, and pistons for an inline, four-cylinder engine.

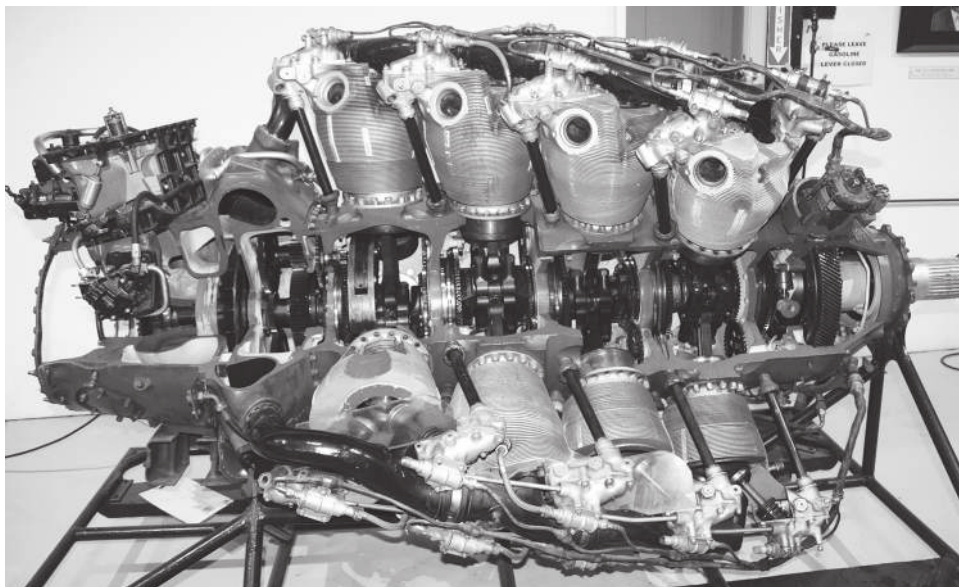
* Used mainly in motor-cycles and boats.



Engine power and torque curves

**FIGURE 14-4**

Cross section of a BMW 5-liter V-12 engine and its power and torque curves *Courtesy of BMW of North America Inc.*

**FIGURE 14-5**

Cutaway view of a WWII vintage, 28-cylinder, Wasp Major radial engine

VEE ENGINES in two-,^{*} four-,^{*} six-, eight-, ten-,[†] and twelve-cylinder[‡] versions are produced, with vee-six and vee-eight being the most common configurations. Figure 14-4 shows a cutaway of a 60° vee-six engine. **Vee engines** can be thought of as *two inline engines grafted together onto a common crankshaft*. The two “inline” portions, or **banks**, are arranged with some **vee angle** between them. Figure 14-1d shows a vee-eight engine. Its crank throws are at 0, 90, 270, and 180° respectively. A vee-eight’s vee angle is 90°. The geometric arrangements of the crankshaft (phase angles) and cylinders (vee angle) have a significant effect on the dynamic condition of the engine. We will soon explore these relationships in detail.

OPPOSED ENGINES are essentially vee engines with a vee angle of 180°. The pistons in each bank are on opposite sides of the crankshaft as shown in Figure 14-6. This arrangement promotes cancellation of inertial forces and is popular in aircraft engines.[§] It has also been used in some automobile and motorcycle applications.^{||}

RADIAL ENGINES have their cylinders arranged radially around the crankshaft in nearly a common plane. These were common on World War II vintage aircraft as they allowed large displacements, and thus high power, in a compact form whose shape was well suited to that of an airplane. Typically air-cooled, the cylinder arrangement allowed good exposure of all cylinders to the airstream. Large versions had multiple rows of radial cylinders, rotationally staggered to allow cooling air to reach the back rows. The gas turbine jet engine has rendered these radial aircraft engines obsolete. Figure 14-5 shows a cutaway view of a 28-cylinder radial engine.

ROTARY ENGINES (Figure 14-7) were an interesting variant on the aircraft radial engine and were used in World War I airplanes.^Δ Although they were similar in appearance and cylinder arrangement to the radial engine, the anomaly was that the **crankshaft** was the stationary ground plane. The propeller was attached to the crankcase (engine block), which rotated around the stationary crankshaft! It is a kinematic inversion of the slider-crank. (See Figure 2-15b.) One advantage is that the piston mass centers can be in pure rotation and so do not impart any vibration to the airframe. All seven connecting rods and pistons are in the same plane. One connecting rod (the “mother” rod) pivots on the crank pin and carries six “daughter” rods on it, as in the radial engine. At least it didn’t need a flywheel.

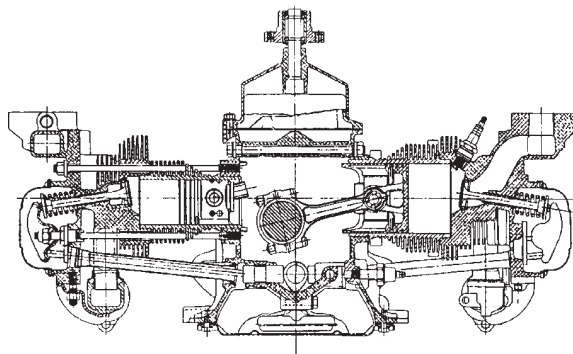


FIGURE 14-6

Chevrolet Corvair horizontally opposed six-cylinder engine

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* Mainly in motorcycles, and boats.

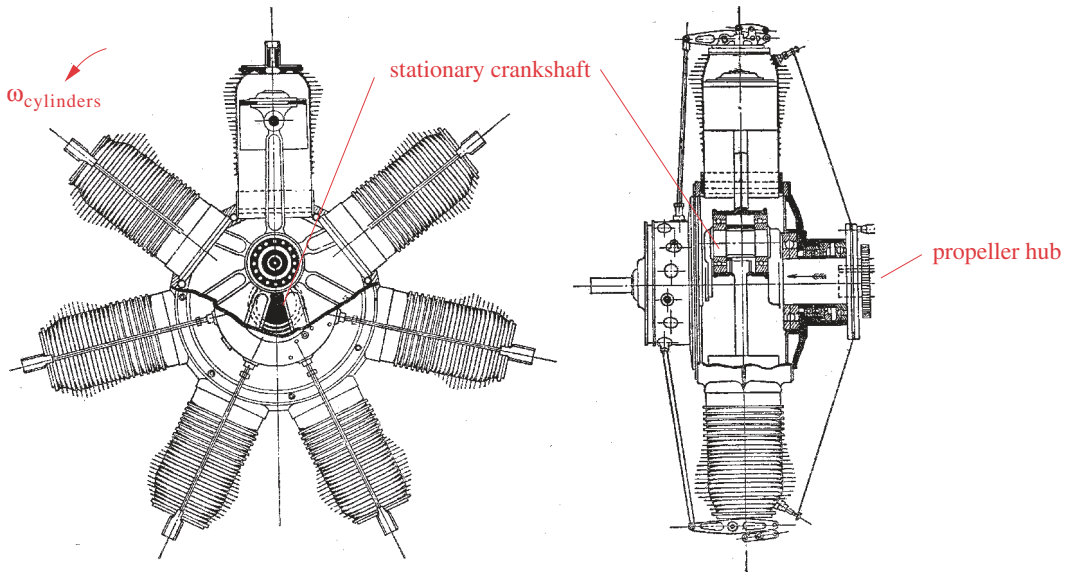
† Honda, Chrysler, Ford, Porsche.

‡ BMW, Jaguar, Mercedes.

§ Continental six-cylinder aircraft engine.

|| Original VW “Beetle” four cylinder, Subaru four and six, Honda motorcycle four and six, Ferrari twelve, Porsche six, the ill-fated Corvair six, and the short-lived Tucker (Continental) six, among others.

Δ Lubrication in the rotary engine was a problem. So-called “loss lubrication” was used, meaning that the lubricant (castor oil) passed through the engine and out the exhaust. This somewhat limited the time that the pilot in his open cockpit could endure inhaling the exhaust. The white silk scarf popular with these pilots was not used for warmth, but rather was placed over mouth and nose to reduce the ingestion of castor oil which is a laxative.

**FIGURE 14-7**

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The Gnome rotary engine (circa 1915). Note the multiple connecting rods on the single stationary crank pin.[1]

Many other configurations of engines have been tried over the century of development of this ubiquitous device. The bibliography at the end of this chapter contains several references which describe other engine designs, the usual, unusual, and exotic. We will begin our detailed exploration of multicylinder engine design with the simplest configuration, the inline engine, and then progress to the vee and opposed versions.

14.2 THE CRANK PHASE DIAGRAM

Fundamental to the design of any multicylinder engine (or piston pump) is the arrangement of crank throws on the crankshaft. We will use the four-cylinder inline engine as an example. Many choices are possible for the crank phase angles in the four-cylinder engine. We will start, for example, with the one that seems most obvious from a commonsense standpoint. There are 360° in any crankshaft. We have four cylinders, so an arrangement of $0, 90, 180,$ and 270° seems appropriate. The **delta phase angle** $\Delta\phi$ between throws is then 90° . In general, for maximum cancellation of inertia forces, which have a period of one revolution, the optimum delta phase angle will be:

$$\Delta\phi_{\text{inertia}} = \frac{360^\circ}{n} \quad (14.1)$$

where n is the number of cylinders.

We must establish some convention for the measurement of these phase angles which will be:

- 1 The first (front) cylinder will be number 1 and its phase angle will always be zero. It is the reference cylinder for all others.

- 2 The phase angles of all other cylinders will be measured with respect to the crank throw for cylinder 1.
- 3 Phase angles are measured internal to the crankshaft, that is, with respect to a rotating coordinate system embedded in the first crank throw.
- 4 Cylinders will be numbered consecutively from front to back of the engine.

The phase angles are defined in a **crank phase diagram** as shown in Figure 14-8 for a four-cylinder, inline engine. Figure 14-8a shows the crankshaft with the throws numbered clockwise around the axis. The shaft is rotating counterclockwise. The pistons are oscillating horizontally in this diagram, along the x axis. Cylinder 1 is shown with its piston at top dead center (TDC). Taking that position as the starting point for the abscissas (thus time zero) in Figure 14-8b, we plot the velocity of each piston for two revolutions of the crank (to accommodate one complete four-stroke cycle). Piston 2 arrives at TDC 90° after piston 1 has left. Thus we say that cylinder 2 lags cylinder 1 by 90 degrees. By convention a *lagging event is defined as having a negative phase angle*, shown by the clockwise numbering of the crank throws. The velocity plots clearly show that each cylinder arrives at TDC (zero velocity) 90° later than the one before it. Negative velocity on the plots in Figure 14-8b indicates piston motion to the left (downstroke) in Figure 14-8a; positive velocity indicates motion to the right (up stroke).

For the discussion in this chapter we will assume counterclockwise rotation of all crankshafts, and all phase angles will thus be negative. We will, however, omit the negative signs on the listings of phase angles with the understanding that they follow this convention.

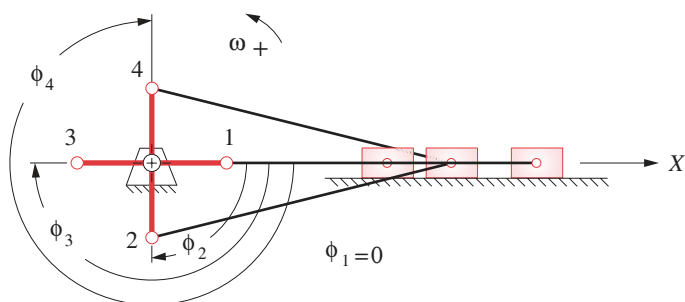
Figure 14-8 shows the timing of events in the cycle and is a necessary and useful aid in defining our crankshaft design. However, it is not necessary to go to the trouble of drawing the correct sinusoidal shapes of the velocity plots to obtain the needed information. All that is needed is a schematic indication of the relative positions within the cycle of the ups and downs of the various cylinders. This same information is conveyed by the simplified crank phase diagram shown in Figure 14-9. Here the piston motions are represented by rectangular blocks with a negative block arbitrarily used to denote a piston downstroke and a positive one a piston upstroke. It is strictly schematic. The positive and negative values of the blocks imply nothing more than that stated. Such a schematic crank phase diagram can (and should) be drawn for any proposed arrangement of crankshaft phase angles. To draw it, simply shift each cylinder's blocks to the right by its phase angle with respect to the first cylinder.

14.3 SHAKING FORCES IN INLINE ENGINES

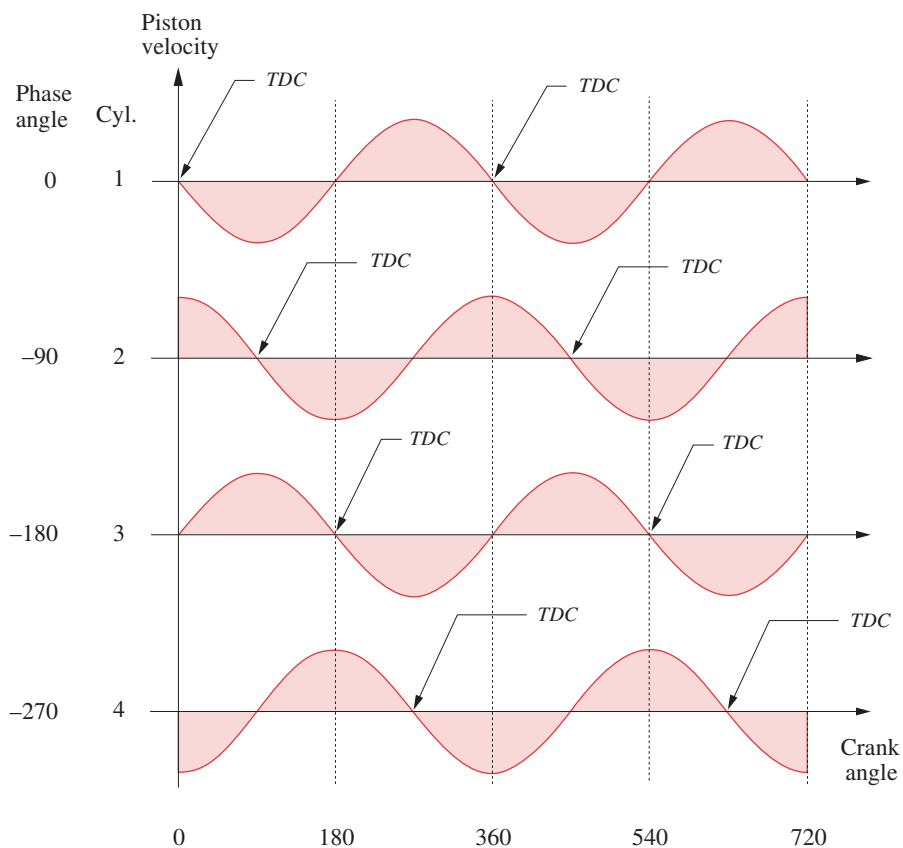
We want to determine the overall shaking force which results from our chosen crankshaft phase angle arrangement. The individual cylinders will each contribute to the total shaking force. We can superpose their effects, taking their phase shifts into account. Equation 13.14e defined the shaking force for one cylinder with the crankshaft rotating at constant ω .

For $a=0$:

$$\mathbf{F}_s \cong \left[m_A r \omega^2 \cos \omega t + m_B r \omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \right] \hat{\mathbf{i}} + \left[m_A r \omega^2 \sin \omega t \right] \hat{\mathbf{j}} \quad (14.2a)$$



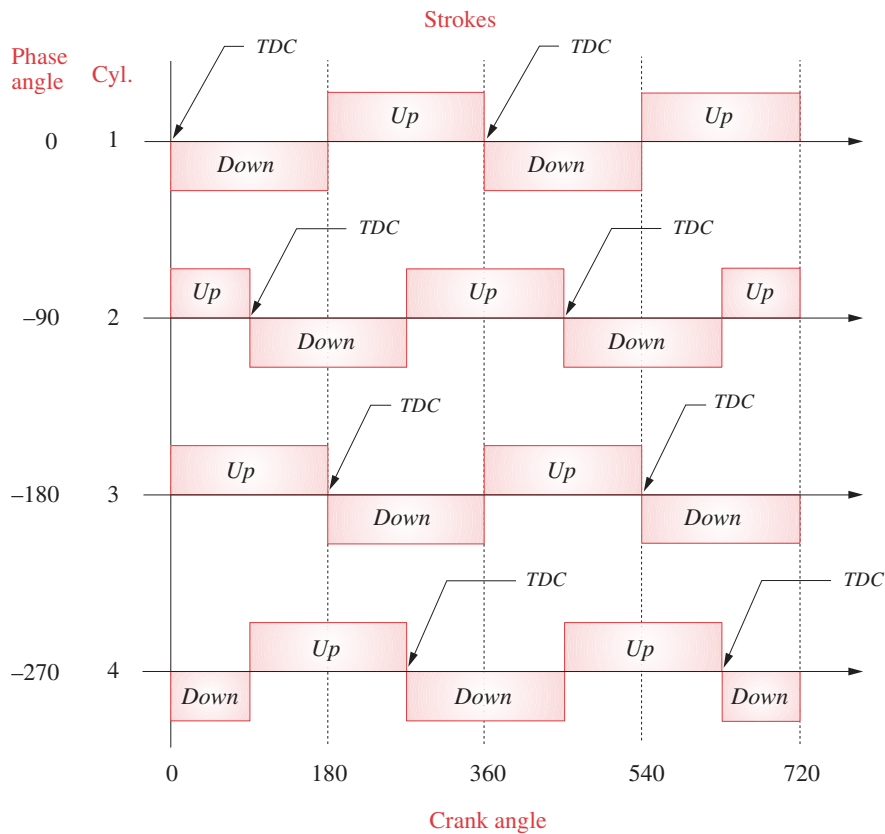
(a) Crankshaft phase angles



(b) The crank phase diagram

FIGURE 14-8

Crank phase angles and the phase diagram

**FIGURE 14-9**

The schematic crank phase diagram

This expression is for an unbalanced crank. In multicylinder engines each crank throw on the crankshaft is at least counterweighted to eliminate the shaking force effects of the combined mass m_A of crank and conrod assumed concentrated at the crank pin. (See Section 13.10 and equation 13.24.) The need for overbalancing is less if the crankshaft phase angles are arranged to cancel the effects of the reciprocating masses at the wrist pins. This inherent balance is possible in inline engines of three or more cylinders, but not in some two-cylinder inline engines. Sometimes the crank throws in an inherently balanced multicylinder engine are also overbalanced in order to reduce the main pin bearing forces as was described in Section 13.10.*

If we provide balance masses with an mr product equal to $m_A r_A$ on each crank throw as shown in Figure 14-3, the terms in equation 14.2a which include m_A will be eliminated, reducing it to:

$$\mathbf{F}_s \approx m_B r \omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \hat{\mathbf{i}} \quad (14.2b)$$

Recall that these are approximate expressions that exclude all harmonics above the second and also assume that each crank throw is exactly balanced, not under- or overbalanced.

* A 90° vee-eight engine typically has approximately $m_B/2$ of overbalance mass added per crank throw to reduce the main pin bearing forces.

We will assume that all cylinders in the engine are of equal displacement and that all pistons and all conrods are interchangeable. This is desirable both for dynamic balance and for lower production costs. If we let the crank angle ωt represent the instantaneous position of the reference crank throw for cylinder 1, the corresponding positions of the other cranks can be defined by their phase angles as shown in Figure 14-8. The total shaking force for a multicylinder inline engine is then:*

* The effect of overbalancing the crank throws is not included in equations 14.2c and 14.2d as shown. The crankshaft is assumed to be exactly balanced here. See Appendix G for the complete equations that include the effects of crank overbalance. Program LINKAGES uses the equations from Appendix G to account for the effects of overbalancing in multicylinder engines.

$$\mathbf{F}_s \equiv m_B r \omega^2 \sum_{i=1}^n \left[\cos(\omega t - \phi_i) + \frac{r}{l} \cos 2(\omega t - \phi_i) \right] \hat{\mathbf{i}} \quad (14.2c)$$

where n = number of cylinders and $\phi_1 = 0$. Substitute the identity:

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

and factor to:

$$\mathbf{F}_s \equiv m_B r \omega^2 \left[\cos \omega t \sum_{i=1}^n \cos \phi_i + \sin \omega t \sum_{i=1}^n \sin \phi_i + \frac{r}{l} \left(\cos 2\omega t \sum_{i=1}^n \cos 2\phi_i + \sin 2\omega t \sum_{i=1}^n \sin 2\phi_i \right) \right] \hat{\mathbf{i}} \quad (14.2d)$$

The ideal value for the shaking force is zero. This expression can only be zero for all values of ωt if:

$$\sum_{i=1}^n \cos \phi_i = 0 \quad \sum_{i=1}^n \sin \phi_i = 0 \quad (14.3a)$$

$$\sum_{i=1}^n \cos 2\phi_i = 0 \quad \sum_{i=1}^n \sin 2\phi_i = 0 \quad (14.3b)$$

Thus, there are some combinations of phase angles ϕ_i which will cause cancellation of the shaking force through the second harmonic. If we wish to cancel higher harmonics, we could reintroduce those harmonics' terms that were truncated from the Fourier series representation and find that the fourth and sixth harmonics will be cancelled if:

$$\sum_{i=1}^n \cos 4\phi_i = 0 \quad \sum_{i=1}^n \sin 4\phi_i = 0 \quad (14.3c)$$

$$\sum_{i=1}^n \cos 6\phi_i = 0 \quad \sum_{i=1}^n \sin 6\phi_i = 0 \quad (14.3d)$$

Equations 14.3 provide us with a convenient predictor of the shaking force behavior of any proposed inline engine design. Program LINKAGES calculates equations 14.3a and 14.3b and displays a table of their values. Note that **both the sine and cosine summations of any multiple of the phase angles must be zero for that harmonic of the shaking force to be zero**. The calculation for our example of a four-cylinder engine with phase angles of $\phi_1 = 0$, $\phi_2 = 90^\circ$, $\phi_3 = 180^\circ$, $\phi_4 = 270^\circ$ in Table 14-1 shows that the shaking forces are zero for the first, second, and sixth harmonics and nonzero for the fourth. So, our commonsense choice in this instance has proven a good one as far as shaking forces

TABLE 14-1 Force Balance State of a 4-Cylinder Inline Engine with a 0, 90, 180, 270° Crankshaft

Primary forces:	$\sum_{i=1}^n \sin \phi_i = 0$	$\sum_{i=1}^n \cos \phi_i = 0$
Secondary forces:	$\sum_{i=1}^n \sin 2\phi_i = 0$	$\sum_{i=1}^n \cos 2\phi_i = 0$
Fourth harmonic forces:	$\sum_{i=1}^n \sin 4\phi_i = 0$	$\sum_{i=1}^n \cos 4\phi_i = 0$
Sixth harmonic forces :	$\sum_{i=1}^n \sin 6\phi_i = 0$	$\sum_{i=1}^n \cos 6\phi_i = 0$

are concerned. As was shown in equation 13.2f, the coefficients of the fourth and sixth harmonic terms are minuscule, so their contributions, if any, can be ignored. The primary component is of most concern, because of its potential magnitude. The secondary (second harmonic) term is less critical than the primary as it is multiplied by r/l which is generally less than 1/3. An unbalanced secondary harmonic of shaking force is undesirable but can be lived with, especially if the engine is of small displacement (less than about 1/2 liter per cylinder).

To see more details on the results of this 0, 90, 180, 270° inline four-cylinder engine configuration, run program LINKAGES, select the configuration from the *Example* pulldown menu, and then *Plot* the shaking force. See Appendix A for more detailed information on program LINKAGES.

14.4 INERTIA TORQUE IN INLINE ENGINES

The inertia torque for a single-cylinder engine was defined in Section 13.6 and equation 13.15e. We are concerned with reducing this inertia torque, preferably to zero, because it combines with the gas torque to form the total torque. (See Section 13.7.) Inertia torque adds nothing to the net driving torque as its average value is always zero, but it does create large oscillations in the total torque which detracts from its smoothness. Inertia torque oscillations can be masked to a degree with the addition of a sufficient flywheel to the system, or their external, net effect can be cancelled by the proper choice of phase angles. However, the torque oscillations, even if hidden from the outside observer, or made to sum to zero, are still present within the crankshaft and can lead to torsional fatigue failure if the part is not properly designed. (See also Figure 14-23.) The approximate one-cylinder inertia torque equation for three harmonics is:

$$\mathbf{T}_{i21} \doteq \frac{1}{2} m_B r^2 \omega^2 \left(\frac{r}{2l} \sin \omega t - \sin 2\omega t - \frac{3r}{2l} \sin 3\omega t \right) \hat{\mathbf{k}} \quad (14.4a)$$

Summing for all cylinders and including their phase angles:

$$\mathbf{T}_{i21} \doteq \frac{1}{2} m_B r^2 \omega^2 \sum_{i=1}^n \left[\frac{r}{2l} \sin(\omega t - \phi_i) - \sin 2(\omega t - \phi_i) - \frac{3r}{2l} \sin 3(\omega t - \phi_i) \right] \hat{\mathbf{k}} \quad (14.4b)$$

Substitute the identity:

$$\sin(a - b) = \sin a \cos b - \cos a \sin b$$

and factor to:

$$\mathbf{T}_{i21} \doteq \frac{1}{2} m_B r^2 \omega^2 \left[\begin{array}{c} \frac{r}{2l} \left(\sin \omega t \sum_{i=1}^n \cos \phi_i - \cos \omega t \sum_{i=1}^n \sin \phi_i \right) \\ - \left(\sin 2\omega t \sum_{i=1}^n \cos 2\phi_i - \cos 2\omega t \sum_{i=1}^n \sin 2\phi_i \right) \\ - \frac{3r}{2l} \left(\sin 3\omega t \sum_{i=1}^n \cos 3\phi_i - \cos 3\omega t \sum_{i=1}^n \sin 3\phi_i \right) \end{array} \right] \hat{\mathbf{k}} \quad (14.4c)$$

This can only be zero for all values of ωt if:

$$\sum_{i=1}^n \sin \phi_i = 0 \quad \sum_{i=1}^n \cos \phi_i = 0 \quad (14.5a)$$

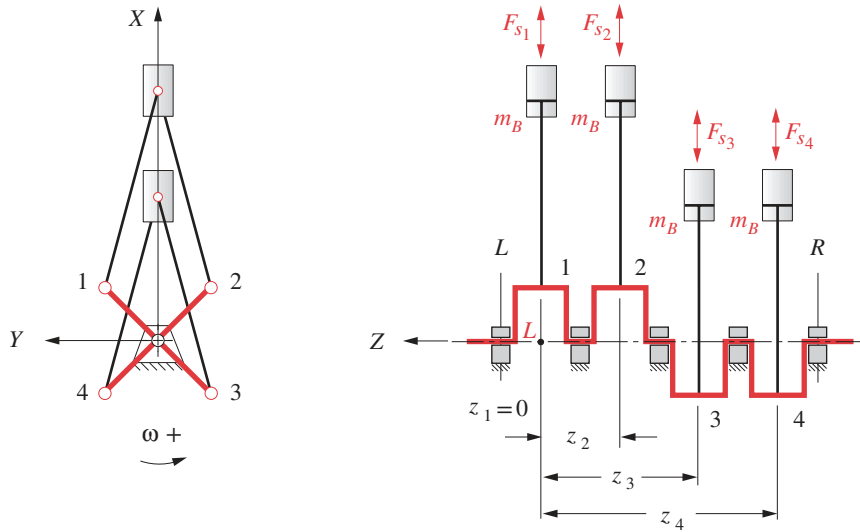
$$\sum_{i=1}^n \sin 2\phi_i = 0 \quad \sum_{i=1}^n \cos 2\phi_i = 0 \quad (14.5b)$$

$$\sum_{i=1}^n \sin 3\phi_i = 0 \quad \sum_{i=1}^n \cos 3\phi_i = 0 \quad (14.5c)$$

Equations 14.5 provide us with a convenient predictor of the inertia torque behavior of any proposed inline engine design. Calculation for our example of a four-cylinder engine with phase angles of $\phi_1 = 0$, $\phi_2 = 90$, $\phi_3 = 180$, $\phi_4 = 270^\circ$ shows that the inertia torque components are zero for the first, second, and third harmonics. So, our current example is a good one for inertia torques as well.

14.5 SHAKING MOMENT IN INLINE ENGINES

We were able to consider the single-cylinder engine to be a single-plane, or two-dimensional, device and thus could statically balance it. The multicylinder engine is three-dimensional. Its multiple cylinders are distributed along the axis of the crankshaft. Even though we may have cancellation of the shaking forces, there may still be unbalanced moments in the plane of the engine block. We need to apply the criteria for dynamic balance. (See Section 12.2 and equation 12.3.) Figure 14-10 shows a schematic of an inline four-cylinder engine with crank phase angles of $\phi_1 = 0$, $\phi_2 = 90$, $\phi_3 = 180$, $\phi_4 = 270^\circ$. The spacing between the cylinders is normally uniform. We can sum moments in the plane of the cylinders about any convenient point such as L at the centerline of the first cylinder:


FIGURE 14-10

Moment arms of the shaking moment

$$\sum M_L = \sum_{i=1}^n z_i F_{s_i} \hat{\mathbf{j}} \quad (14.6a)$$

where $\mathbf{F}_{s_i}$ is the shaking force and z_i is the moment arm of the i th cylinder.* Substituting equation 14.2d for $\mathbf{F}_{s_i}$:

$$\sum M_L \equiv m_B r \omega^2 \left[\cos \omega t \sum_{i=1}^n z_i \cos \phi_i + \sin \omega t \sum_{i=1}^n z_i \sin \phi_i + \frac{r}{l} \left(\cos 2\omega t \sum_{i=1}^n z_i \cos 2\phi_i + \sin 2\omega t \sum_{i=1}^n z_i \sin 2\phi_i \right) \right] \hat{\mathbf{j}} \quad (14.6b)$$

This expression can only be zero for all values of ωt if:

$$\sum_{i=1}^n z_i \cos \phi_i = 0 \quad \sum_{i=1}^n z_i \sin \phi_i = 0 \quad (14.7a)$$

$$\sum_{i=1}^n z_i \cos 2\phi_i = 0 \quad \sum_{i=1}^n z_i \sin 2\phi_i = 0 \quad (14.7b)$$

These will guarantee no shaking moments through the second harmonic. We can extend this to higher harmonics as we did for the shaking force.

$$\sum_{i=1}^n z_i \cos 4\phi_i = 0 \quad \sum_{i=1}^n z_i \sin 4\phi_i = 0 \quad (14.7c)$$

$$\sum_{i=1}^n z_i \cos 6\phi_i = 0 \quad \sum_{i=1}^n z_i \sin 6\phi_i = 0 \quad (14.7d)$$

* The effect of overbalancing the crank throws is not included in equation 14.6b as shown. The crankshaft is assumed to be exactly balanced here. See Appendix G for the complete equations that include the effects of crank overbalance.

Program LINKAGES uses the equations from Appendix G to account for the effects of overbalancing in multicylinder engines.

TABLE 14-2 Moment Balance State of a 4-Cylinder, Inline Engine with a 0, 90, 180, 270° Crankshaft, and $z_1 = 0$, $z_2 = 1$, $z_3 = 2$, $z_4 = 3$

Primary moments:	$\sum_{i=1}^n z_i \sin \phi_i = -2$	$\sum_{i=1}^n z_i \cos \phi_i = -2$
Secondary moments:	$\sum_{i=1}^n z_i \sin 2\phi_i = 0$	$\sum_{i=1}^n z_i \cos 2\phi_i = -2$

Note that both the sine and cosine summations of any multiple of the phase angles must be zero for that harmonic of the shaking moment to be zero. The calculation for our example of a four-cylinder engine with phase angles of $\phi_1 = 0$, $\phi_2 = 90$, $\phi_3 = 180$, $\phi_4 = 270^\circ$, and an assumed cylinder spacing of one length unit ($z_2 = 1$, $z_3 = 2$, $z_4 = 3$) in Table 14-2, shows that the shaking moments are not zero for any of these harmonics. So, our choice of phase angles, which is a good one for shaking forces and torques, fails the test for zero shaking moments. The coefficients of the fourth and sixth harmonic terms in the moment equations are minuscule, so they will be ignored. The secondary (second harmonic) term is less critical than the primary as it is multiplied by r/l which is generally less than 1/3. An unbalanced secondary harmonic of shaking moment is undesirable but can be tolerated, especially if the engine is of small displacement (less than about 1/2 liter per cylinder). The primary component is of greatest concern, because of its magnitude. If we wish to use this crankshaft configuration, we will need to apply a balancing technique to the engine as described in a later section to at least eliminate the primary moment. A large shaking moment is undesirable as it will cause the engine to **pitch** forward and back (like a bucking bronco) as the moment oscillates from positive to negative in the plane of the cylinders. *Do not confuse this shaking moment with the shaking torque* which acts to **roll** the engine back and forth about the Z axis of the crankshaft.

Figure 14-11 shows the primary and secondary components of the shaking moment for this example engine for two revolutions of the crank. Each is a pure harmonic of zero average value. The total moment is the sum of these two components. This engine configuration is a built-in example in program LINKAGES. See Appendix A for information about the program.

* http://www.designof-machinery.com/DOM/Even_Firing.mp4

14.6 EVEN FIRING *Watch a Video on Even Firing (47:29)**

The inertial forces, torques, and moments are only one set of criteria which need to be considered in the design of multicylinder engines. Gas force and gas torque considerations are equally important. In general, it is desirable to create a firing pattern among the cylinders that is evenly spaced in time. If the cylinders fire unevenly, vibrations will be created which may be unacceptable. Smoothness of the power pulses is desired. The power pulses depend on the stroke cycle. If the engine is a two-stroke, there will be one power pulse per revolution in each of its n cylinders. The optimum delta phase angle between the cylinders' crank throws for evenly spaced power pulses will then be:

$$\Delta \phi_{two\ stroke} = \frac{360^\circ}{n} \quad (14.8a)$$

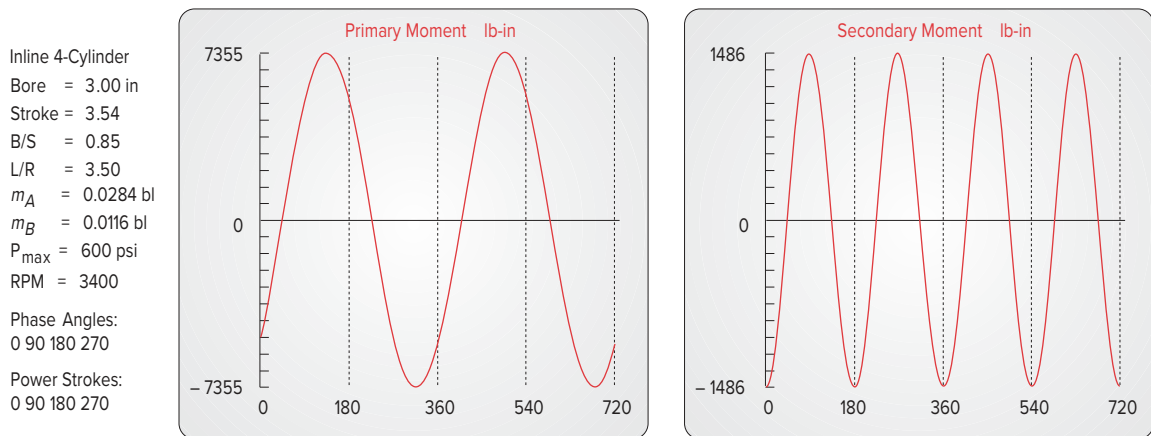


FIGURE 14-11

Primary and secondary moments in the 0, 90, 180, 270° crankshaft four-cylinder engine

For a four-stroke engine there will be one power pulse in each cylinder every two revolutions. The optimum delta phase angle of the crank throws for evenly spaced power pulses will then be:

$$\Delta\phi_{four\ stroke} = \frac{720^\circ}{n} \quad (14.8b)$$

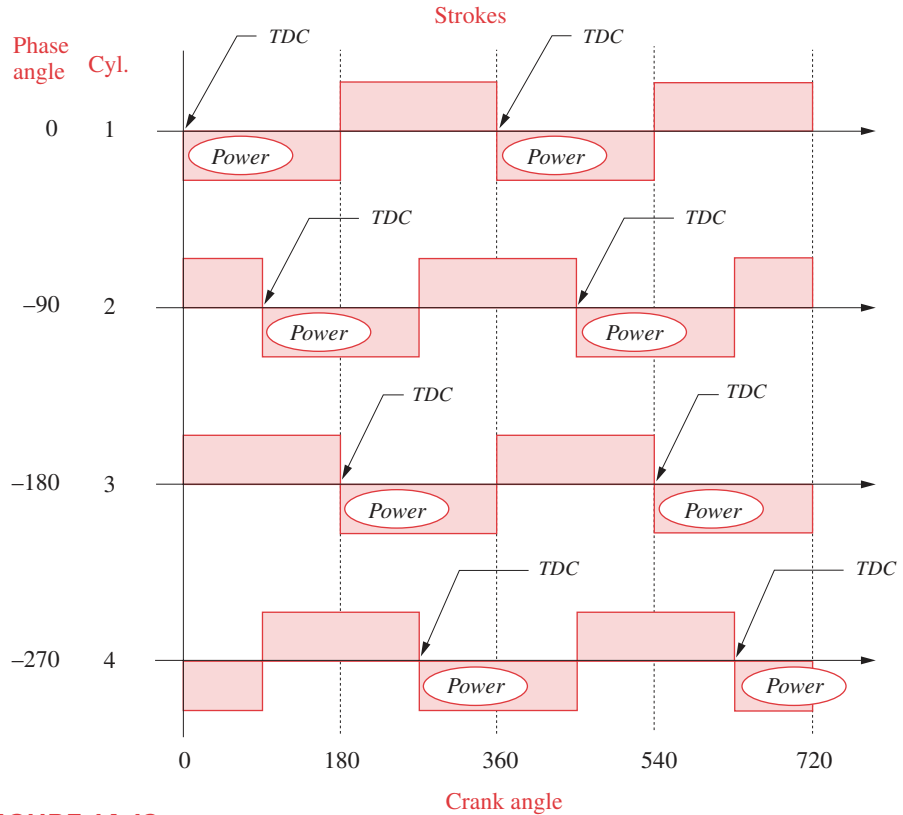
Compare equations 14.8a and 14.8b to equation 14.1 which defined the optimum delta phase angle for cancellation of inertia forces. A two-stroke engine can have both even firing and inertia balance, but a four-stroke engine has a conflict between these two criteria. Thus some design trade-offs will be necessary to obtain the best compromise between these factors in the four-stroke case.

Two-Stroke Cycle Engine

To determine the firing pattern of an engine design, we must return to the crank phase diagram. Figure 14-12 reproduces Figure 14-9 and adds new information to it. It shows the power pulses for a **two-stroke cycle, four-cylinder engine** with the $\phi_i = 0, 90, 180, 270^\circ$ phase angle crank configuration. Note that each cylinder's negative block in Figure 14-12 is shifted to the right by its phase angle with respect to reference cylinder 1. In this schematic representation, only the negative blocks on the diagram are available for power pulses as they represent the downstroke of the piston. By convention, cylinder 1 fires first, so its negative block at 0° is labeled **Power**. The other cylinders

may be fired in any order, but their power pulses should be as evenly spaced as possible across the interval.

The available power pulse spacings are dictated by the crank phase angles. There may be more than one firing order which will give even firing, especially with large numbers of cylinders. In this simple example the firing order 1, 2, 3, 4 will work as it will provide successive power pulses every 90° across the interval. The **power stroke angles** ψ_i are the angles in the cycle at which the cylinders fire. They are defined by the crank-

**FIGURE 14-12**

Two-stroke inline four-cylinder engine crank phase diagram with $\phi_i = 0, 90, 180, 270^\circ$

shaft phase angles and the choice of firing order in combination, and in this example are $\psi_i = 0, 90, 180$, and 270° . In general, ψ_i are not equal to ϕ_i . Their correspondence with the phase angles in this example results from choosing the consecutive firing order 1, 2, 3, 4.

For a **two-stroke engine**, the power stroke angles ψ_i must be *between 0 and 360°*. We always want them to be evenly spaced in that interval with a delta power stroke angle defined by equation 14.8c. For our four-cylinder, two-stroke engine, ideal power stroke angles are then $\psi_i = 0, 90, 180, 270^\circ$, which we have achieved in this example.

We define the **delta power stroke angle** differently for each stroke cycle. For the two-stroke engine:

$$\Delta\psi_{\text{two stroke}} = \frac{360^\circ}{n} \quad (14.8c)$$

For the four-stroke engine:

$$\Delta\psi_{\text{four stroke}} = \frac{720^\circ}{n} \quad (14.8d)$$

The gas torque for a one-cylinder engine was defined in equation 13.8b. The combined gas torque for all cylinders must sum the contributions of n cylinders, each phase-shifted by its power stroke angle ψ_i :

$$\mathbf{T}_{g21} \cong F_g r \sum_{i=1}^n \left\{ \sin(\omega t - \psi_i) \left[1 + \frac{r}{l} \cos(\omega t - \psi_i) \right] \right\} \hat{\mathbf{k}} \quad (14.9)$$

Figure 14-13 shows the gas torque, inertia torque, and shaking force for this two-stroke four-cylinder engine plotted from program LINKAGES. The shaking moment components are shown in Figure 14-11. Except for the unbalanced shaking moments, this design is otherwise acceptable. The inertia force and inertia torque are both zero which is ideal. The gas torque consists of uniformly shaped and spaced pulses across the interval, four per revolution. Note that program LINKAGES plots two full revolutions to accommodate the four-stroke case; thus eight power pulses are seen. Open the file F14-13.eng in the program to exercise this example.

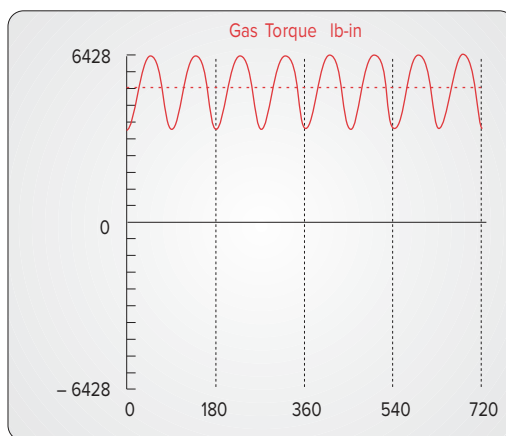
Four-Stroke Cycle Engine

Figure 14-14 shows a crank phase diagram for the *same crankshaft design* as in Figure 14-12 except that it is designed as a *four-stroke cycle engine*. There is now only one power stroke every 720° for each cylinder. The second negative block for each cylinder must be used for the intake stroke. Cylinder 1 is again fired first. An evenly spaced pattern of power pulses among the other cylinders is again desired but is now not possible with this crankshaft. Whether the firing order is 1, 3, 4, 2 or 1, 2, 4, 3 or 1, 4, 2, 3, or any other chosen, there will be both gaps and overlaps in the power pulses. The first firing order listed, 1, 3, 4, 2, has been chosen for this example. This results in the set of power stroke angles $\psi_i = 0, 180, 270, 450^\circ$. These **power stroke angles** define the points in the **720° cycle** where each cylinder fires. Thus for a four-stroke engine, the power stroke angles ψ_i must be between 0 and 720° . We would like them to be evenly spaced in that interval with a delta angle defined by equation 14.8d. For our four-cylinder, four-stroke engine, the ideal power stroke angles would then be $\psi_i = 0, 180, 360, 540^\circ$. We clearly have not achieved them in this example. Figure 14-15 shows the resulting gas torque. Open the file F14-15.eng in program LINKAGES to exercise this example.

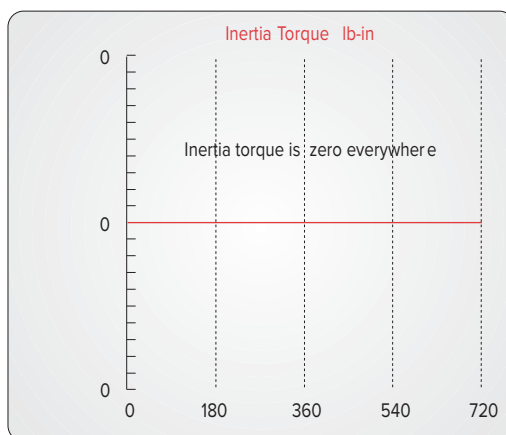
The uneven firing in Figure 14-15 is obvious. This uneven gas torque will be perceived by the operator of any vehicle containing this engine as rough running and vibration, especially at idle speed. At higher engine speeds the flywheel will tend to mask this roughness, but flywheels are ineffective at low speeds. It is this fact that causes most engine designers to *favor even firing over elimination of inertia effects* in their selection of crankshaft phase angles. The inertia force, torque, and moment are all functions of engine speed squared. But, as engine speed increases the magnitude of these factors, the same speed is also increasing the flywheel's ability to mask their effects. Not so with gas-torque roughness due to uneven firing. It is bad at all speeds and the flywheel won't hide it at low speed.

We therefore must reject this crankshaft design for our four-stroke, four-cylinder engine. Equation 14.8b indicates that we need a delta phase angle $\Delta\phi_i = 180^\circ$ in our crankshaft to obtain even firing. We need four crank throws, and all crank phase angles must be less than 360° . So, we must repeat some angles if we use a delta phase angle of

(a)



(b)



Inline 4-Cylinder
2-Stroke Cycle

Bore = 3.00 in

Stroke = 3.54

B/S = 0.85

L/R = 3.50

m_A = 0.0284 bl

m_B = 0.0116 bl

P_{max} = 600 psi

RPM = 3400

Phase Angles:

0 90 180 270

Power Strokes:

0 90 180 270

(c)

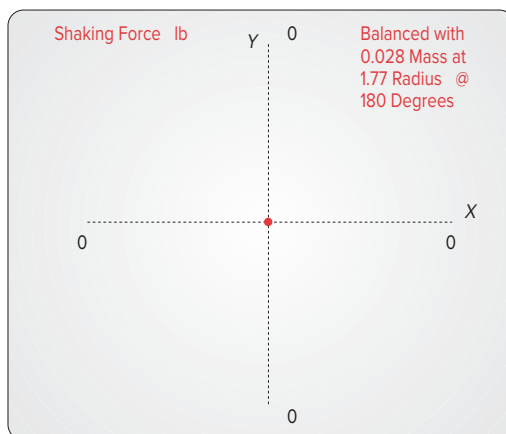


FIGURE 14-13

Torque and shaking force in the two-stroke four-cylinder inline engine

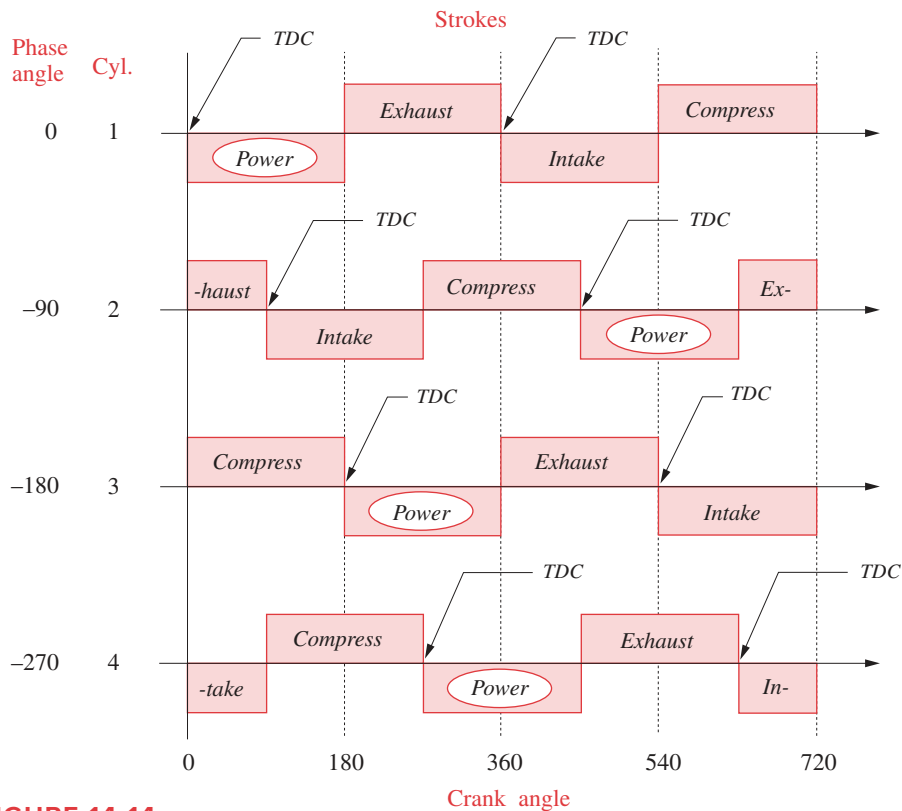


FIGURE 14-14

Four-stroke inline four-cylinder engine crank phase diagram with $\phi_i = 0, 90, 180, 270^\circ$

Inline 4-Cylinder
4-Stroke Cycle
Bore = 3.00 in
Stroke = 3.54
B/S = 0.85
L/R = 3.50
 $m_A = 0.0284$ lb
 $m_B = 0.0116$ lb
 $P_{\max} = 600$ psi
RPM = 3400
Phase Angles:
0 90 180 270
Power Strokes:
0 90 180 270

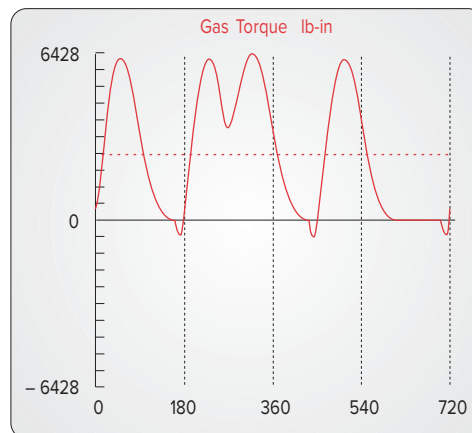
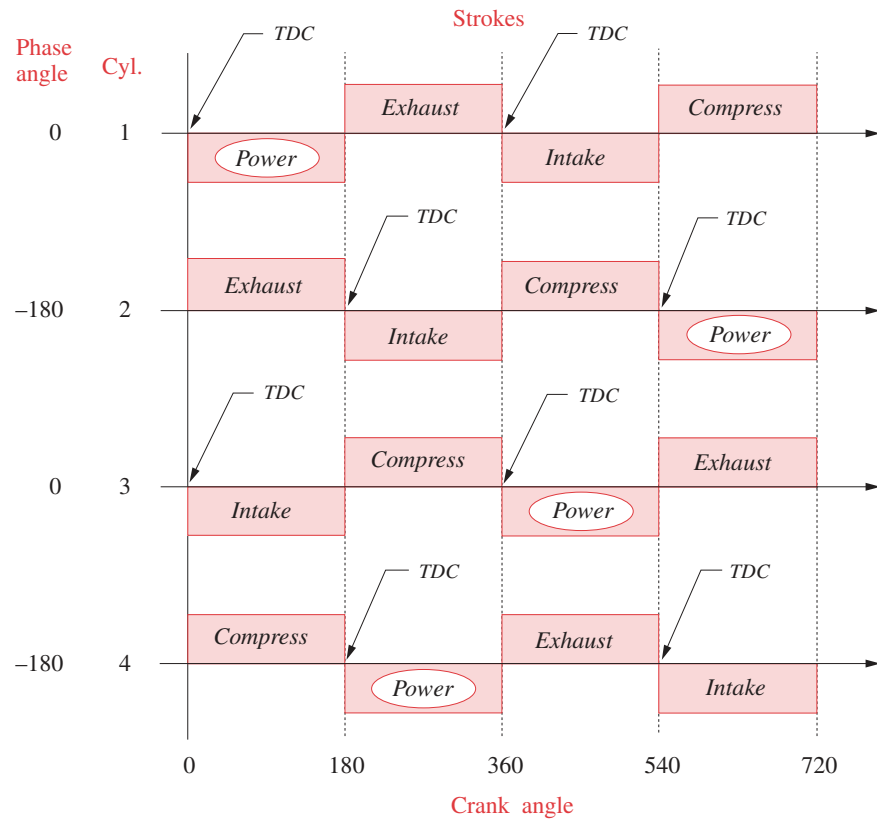


FIGURE 14-15

Uneven firing four-stroke, four-cylinder inline engine with a 0, 90, 180, 270° crankshaft

**FIGURE 14-16**

Even firing four-stroke, four-cylinder engine crank phase diagram with $\phi_i = 0, 180, 0, 180^\circ$

* Note that $0, 180, 360, 540^\circ$, modulo 360 is the same as $0, 180, 0, 180$.

† Note the pattern of acceptable firing orders (FO). Write two revolutions' worth of any acceptable FO, as in 1, 4, 3, 2, 1, 4, 3, 2. Any set of four successive numbers in this sequence, either forward or backward, is an acceptable FO. If we require the first to be cylinder 1, then the only other possibility here is the backward set 1, 2, 3, 4.

180° . One possibility is $\phi_i = 0, 180, 0, 180^\circ$ for the four crank throws.* The crank phase diagram for this design is shown in Figure 14-16. The power strokes can now be evenly spaced over 720° . A firing order of 1, 4, 3, 2 has been chosen which gives the desired sequence of power stroke angles, $\psi_i = 0, 180, 360, 540^\circ$. (Note that a firing order of 1, 2, 3, 4 would also work with this engine.)†

The inertial balance condition of this design must now be checked with equations 14.3, 14.5, and 14.7. These show that the primary inertia force is zero, but the primary moment, secondary force, secondary moment, and inertia torque are all nonzero as shown in Table 14-3. So, this even-firing design has compromised the very good state of inertia balance of the previous design in order to achieve even firing. The inertia torque variations can be masked by a flywheel. The secondary forces and moments are relatively small in a small engine and can be tolerated. The nonzero primary moment is a problem which needs to be addressed. To see the results of this engine configuration, run program LINKAGES and select it from the *Example* pulldown menu. Then plot the results. See Appendix A for more detailed instructions on the use of program LINKAGES.

We shall soon discuss ways to counter an unbalanced moment with the addition of balance shafts, but there is a more direct approach available in this example. Figure 14-17

TABLE 14-3 Force and Moment Balance State of a 4-Cylinder, Inline Engine with a 0, 180, 0, 180° Crankshaft, and $z_1 = 0, z_2 = 1, z_3 = 2, z_4 = 3$

Primary forces:	$\sum_{i=1}^n \sin \phi_i = 0$	$\sum_{i=1}^n \cos \phi_i = 0$
Secondary forces:	$\sum_{i=1}^n \sin 2\phi_i = 0$	$\sum_{i=1}^n \cos 2\phi_i = 4$
Primary moments:	$\sum_{i=1}^n z_i \sin \phi_i = 0$	$\sum_{i=1}^n z_i \cos \phi_i = -2$
Secondary moments:	$\sum_{i=1}^n z_i \sin 2\phi_i = 0$	$\sum_{i=1}^n z_i \cos 2\phi_i = 6$

shows that the shaking moment is due to the action of the individual cylinders' inertial forces acting at moment arms about some center. If we consider that center to be point C

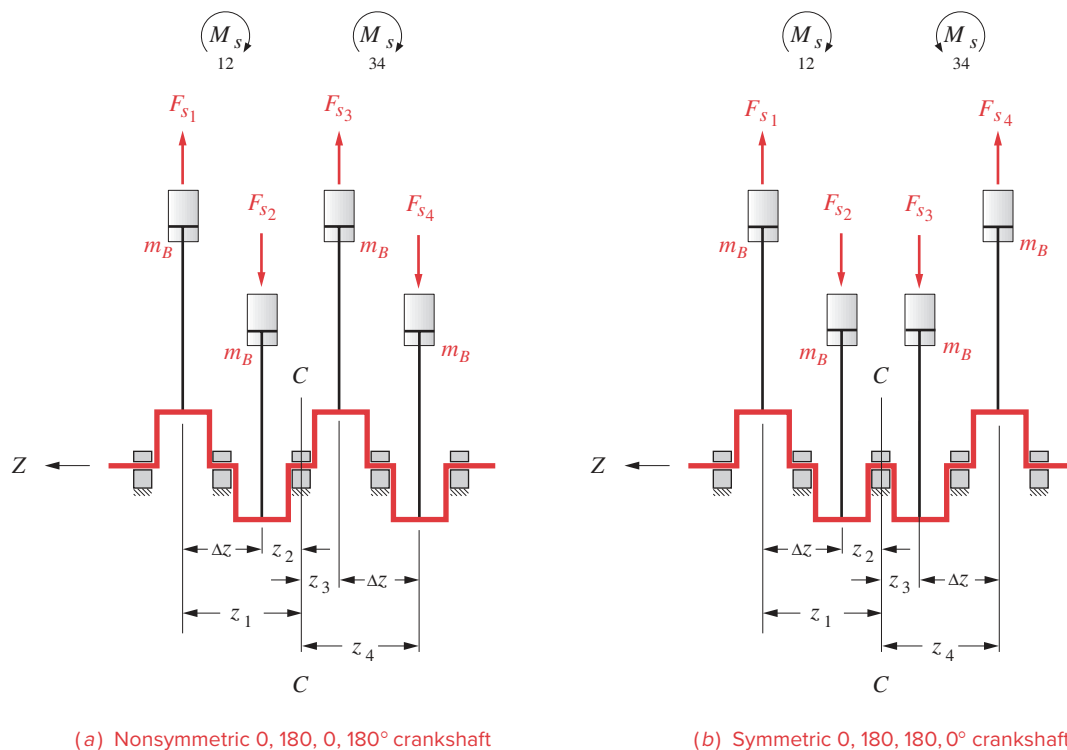


FIGURE 14-17

Mirror-symmetric crankshafts cancel primary moments.

in the middle of the engine, it should be apparent that any primary force-balanced crankshaft design which is mirror-symmetric about a transverse plane through point C would also have balanced primary moments as long as all cylinder spacings were uniform and all inertial forces equal. Figure 14-17a shows the $0, 180, 0, 180^\circ$ crankshaft which is not mirror-symmetric. The couple $\mathbf{F}_{s_1} \Delta z$ due to cylinder pairs 1, 2 has the same magnitude and sense as the couple $\mathbf{F}_{s_3} \Delta z$ due to cylinders 3 and 4, so they add. Figure 14-17b shows the $0, 180, 180, 0^\circ$ crankshaft which is *mirror-symmetric*. The couple $\mathbf{F}_{s_1} \Delta z$ due to cylinder pairs 1, 2 has the same magnitude but opposite sense to the couple $\mathbf{F}_{s_3} \Delta z$ due to cylinders 3, 4, so they cancel. We can then achieve both even firing and balanced primary moments by changing the sequence of crank throw phase angles to $\phi_i = 0, 180, 180, 0^\circ$ which is *mirror-symmetric*.

The crank phase diagram for this design is shown in Figure 14-18. The power strokes can still be evenly spaced over 720° . A firing order of 1, 3, 4, 2 has been chosen which gives the same desired sequence of power stroke angles, $\psi_i = 0, 180, 360, 540^\circ$. (Note that a firing order of 1, 2, 4, 3 would also work with this engine.)* Equations 14.3, 14.5, and 14.7 and Table 14-4 show that the primary inertia force and primary moment are both now zero, but the secondary force, secondary moment, and inertia torque are still nonzero.

* In inline engines, and within any one bank of a vee engine, a nonconsecutive firing order (i.e., not 1, 2, 3, 4) is usually preferred so that adjacent cylinders do not fire sequentially. This allows the intake manifold more time to recharge locally between intake strokes and the exhaust manifold to scavenge more efficiently.

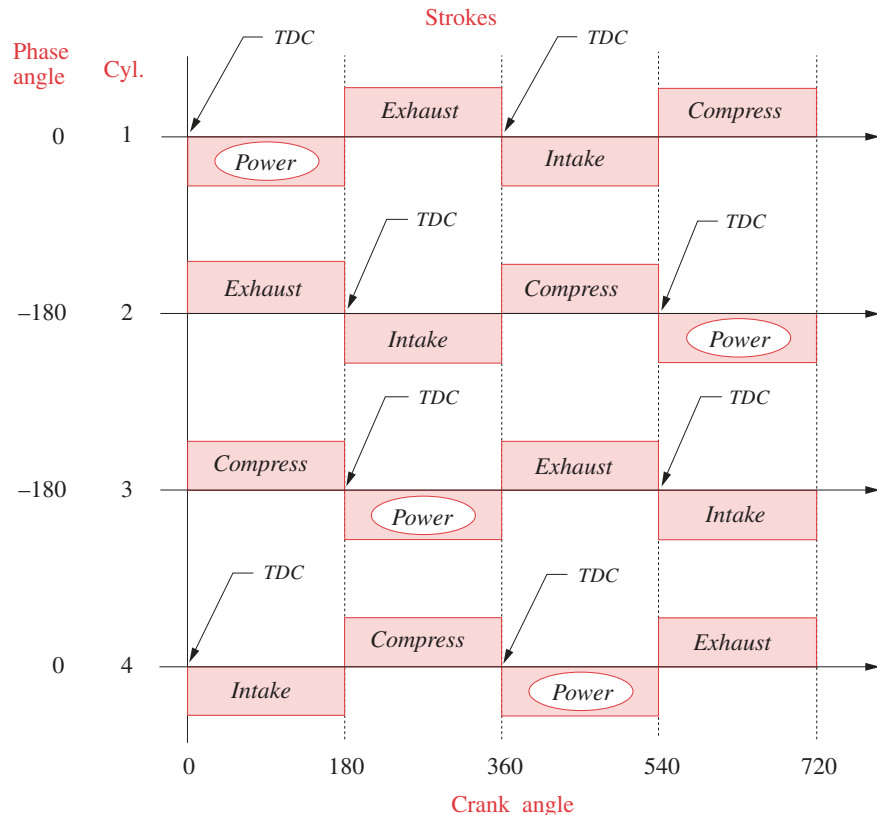


FIGURE 14-18

Even firing four-stroke, four-cylinder engine crank phase diagram with a mirror-symmetric $0, 180, 180, 0^\circ$ crankshaft

TABLE 14-4 Force and Moment Balance State of a 4-Cylinder, Inline Engine with a 0, 180, 180, 0° Crankshaft, and $z_1 = 0$, $z_2 = 1$, $z_3 = 2$, $z_4 = 3$

Primary forces:	$\sum_{i=1}^n \sin \phi_i = 0$	$\sum_{i=1}^n \cos \phi_i = 0$
Secondary forces:	$\sum_{i=1}^n \sin 2\phi_i = 0$	$\sum_{i=1}^n \cos 2\phi_i = 4$
Primary moments:	$\sum_{i=1}^n z_i \sin \phi_i = 0$	$\sum_{i=1}^n z_i \cos \phi_i = 0$
Secondary moments:	$\sum_{i=1}^n z_i \sin 2\phi_i = 0$	$\sum_{i=1}^n z_i \cos 2\phi_i = 6$

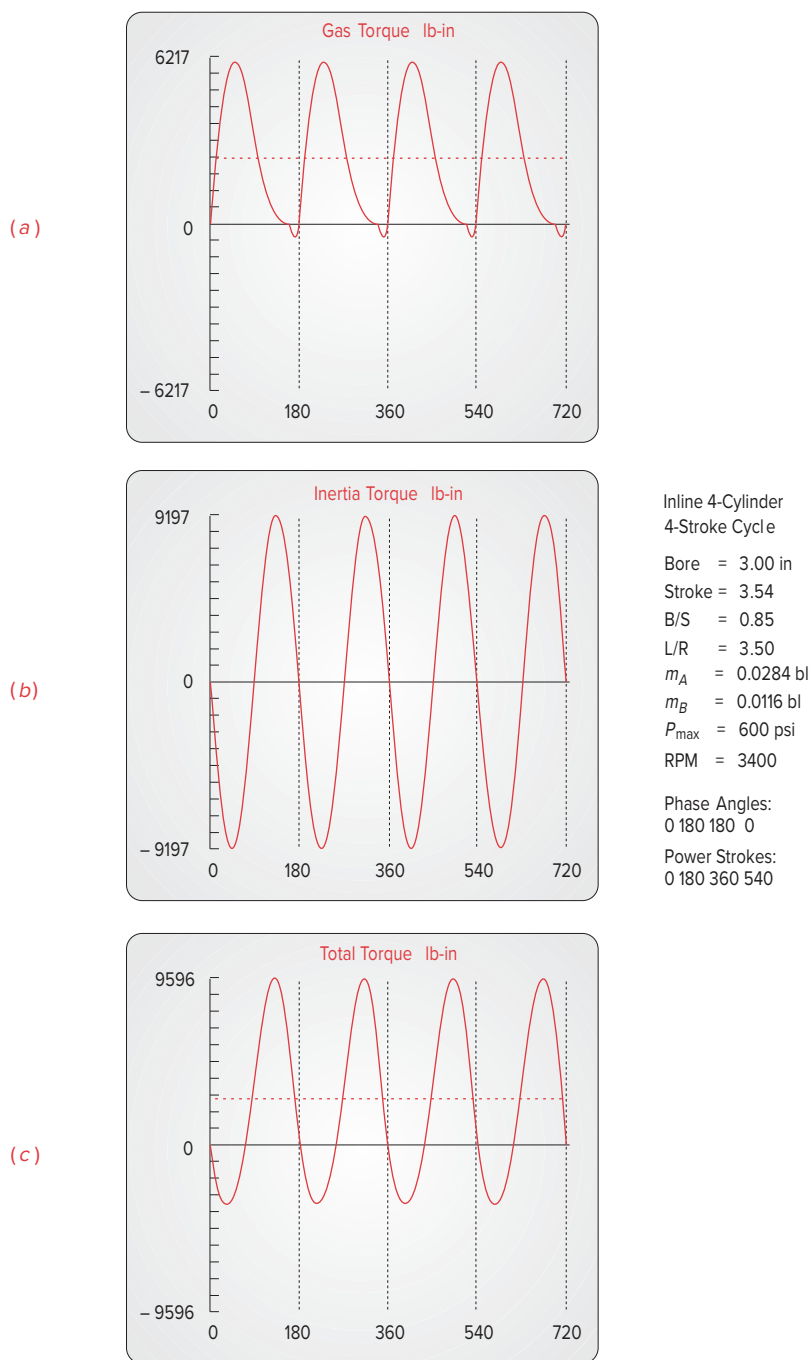
This $\phi_i = 0, 180, 180, 0^\circ$ crankshaft is considered the best design trade-off and is the one universally used in these four-cylinder inline, four-stroke production engines. Figures 14-2 and 14-3 show such a four-cylinder design. Inertia balance is sacrificed to gain even firing for the reasons cited before. Figure 14-19 shows the gas torque, inertia torque, and total torque for this design. Figure 14-20 shows the secondary shaking moment, secondary shaking force component, and a polar plot of the total shaking force for this design. Note that Figures 14-20b and 14-20c are just different views of the same parameter. The polar plot of the shaking force in Figure 14-20c is a view of the shaking force looking at the end of the crankshaft axis with the piston motion horizontal. The cartesian plot in Figure 14-20b shows the same force on a time axis. Since the primary component is zero, this total force is due only to the secondary component. We will soon discuss ways to eliminate these secondary forces and moments.

To see the results of this engine configuration, run program LINKAGES and select it from the *Examples* pulldown menu. Then *Plot* or *Print* the results. See Appendix A for more detailed instructions on the use of the program.

14.7 VEE ENGINES [Watch a Video on Vee Engines \(48:25\)](#)*

The same design principles which apply to inline engines also apply to vee and opposed configurations. Even firing takes precedence over inertia balance and mirror symmetry of the crankshaft balances primary moments. In general, a vee engine will have similar inertia balance to that of the inline engines from which it is constructed. A vee-six is essentially two three-cylinder inline engines on a common crankshaft, a vee-eight is two four-cylinder inlines, etc. The larger number of cylinders allows more power pulses to be spaced out over the cycle for a smoother (and larger average) gas torque. The existence of a **vee angle** between the two inline engines introduces an additional phase shift of the inertial and gas events which is analogous to, but independent of, the phase angle effects. This vee angle is the designer's choice, but there are good and poor choices. The same criteria of even firing and inertia balance apply to its selection.

* http://www.designofmachinery.com/DOM/Vee_Engines.mp4

**FIGURE 14-19**

Torque in the four-stroke, four-cylinder inline engine with a 0, 180, 180, 0° crankshaft

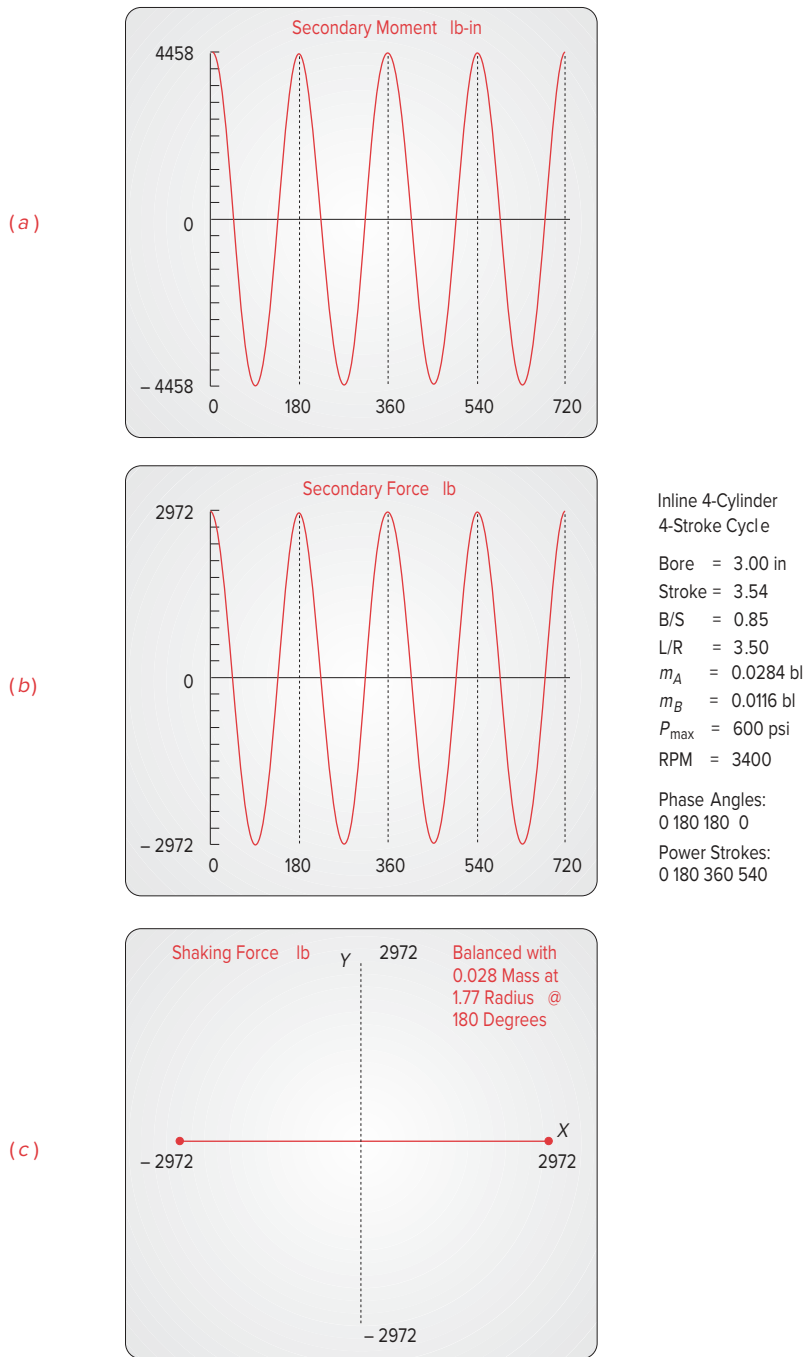


FIGURE 14-20

Shaking forces and moments in the four-stroke, four-cylinder 0, 180, 180, 0° engine

The **vee angle** $v = 2\gamma$ is defined as shown in Figure 14-21. Each bank is offset by its **bank angle** γ referenced to the central X axis of the engine. The crank angle ωt is measured from the X axis. Cylinder 1 in the right bank is the reference cylinder. Events in each bank will now be phase-shifted by its bank angle as well as by the crankshaft phase angles. These two phase shifts will superpose. Taking any one cylinder in either bank as an example, let its instantaneous crank angle be represented by:

$$\theta = \omega t - \phi_i \quad (14.10a)$$

Consider first a two-cylinder vee engine with one cylinder in each bank and with both sharing a common crank throw. The shaking force for a single cylinder in the direction of piston motion $\hat{u}$ with θ measured from the piston axis is:

$$\mathbf{F}_s \doteq m_B r \omega^2 \left(\cos \theta + \frac{r}{l} \cos 2\theta \right) \hat{u} \quad (14.10b)$$

The total shaking force is the vector sum of the contributions from each bank.

$$\mathbf{F}_s = \mathbf{F}_{s_L} + \mathbf{F}_{s_R} \quad (14.10c)$$

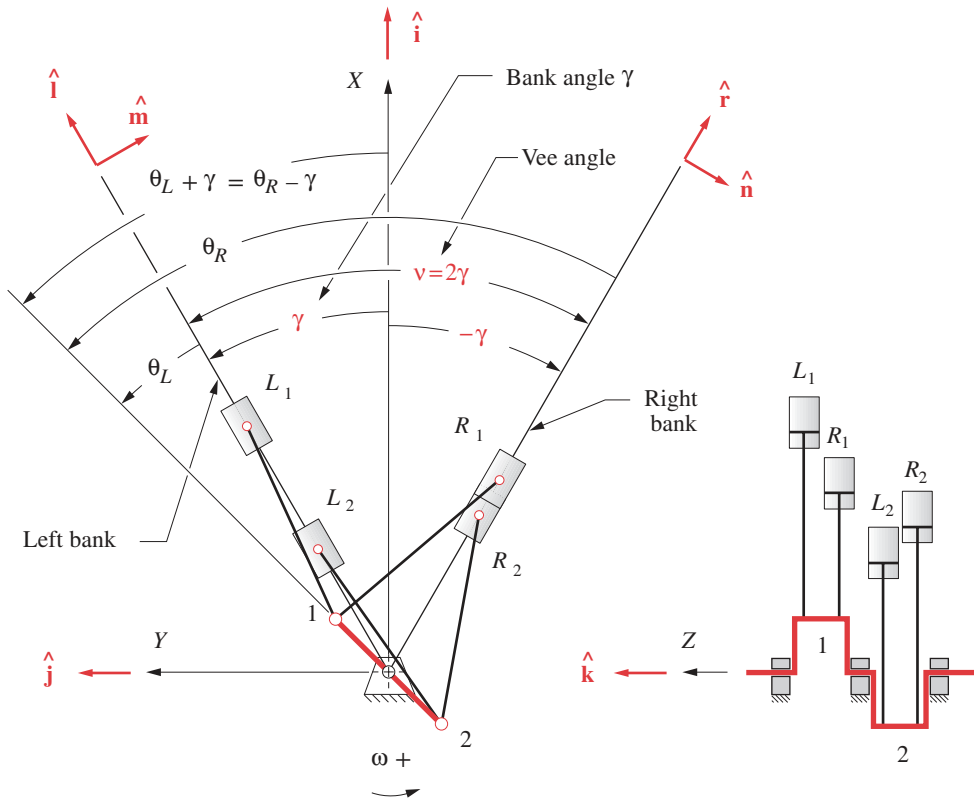


FIGURE 14-21

Vee-engine geometry

We now want to refer the crank angle to the central X axis. The shaking forces for the right (R) and left (L) banks, in the planes of the respective cylinder banks, can then be expressed as:

$$\begin{aligned}\mathbf{F}_{s_R} &\cong m_B r \omega^2 \left[\cos(\theta + \gamma) + \frac{r}{l} \cos 2(\theta + \gamma) \right] \hat{\mathbf{r}} \\ \mathbf{F}_{s_L} &\cong m_B r \omega^2 \left[\cos(\theta - \gamma) + \frac{r}{l} \cos 2(\theta - \gamma) \right] \hat{\mathbf{l}}\end{aligned}\quad (14.10d)$$

Note that the bank angle γ is added to or subtracted from the crank angle for each cylinder bank to reference it to the central X axis. The forces are still directed along the planes of the cylinder banks. Substitute the identities:

$$\begin{aligned}\cos(\theta + \gamma) &= \cos \theta \cos \gamma - \sin \theta \sin \gamma \\ \cos(\theta - \gamma) &= \cos \theta \cos \gamma + \sin \theta \sin \gamma\end{aligned}\quad (14.10e)$$

to get:

$$\begin{aligned}\mathbf{F}_{s_R} &\cong m_B r \omega^2 \left[\begin{aligned} &\cos \theta \cos \gamma - \sin \theta \sin \gamma \\ &+ \frac{r}{l} (\cos 2\theta \cos 2\gamma - \sin 2\theta \sin 2\gamma) \end{aligned} \right] \hat{\mathbf{r}} \\ \mathbf{F}_{s_L} &\cong m_B r \omega^2 \left[\begin{aligned} &\cos \theta \cos \gamma + \sin \theta \sin \gamma \\ &+ \frac{r}{l} (\cos 2\theta \cos 2\gamma + \sin 2\theta \sin 2\gamma) \end{aligned} \right] \hat{\mathbf{l}}\end{aligned}\quad (14.10f)$$

Now, to account for the possibility of multiple cylinders, phase-shifted within each bank, substitute equation 14.10a for θ and replace the sums of angle terms with products from the identities:

$$\begin{aligned}\cos(\omega t - \phi_i) &= \cos \omega t \cos \phi_i + \sin \omega t \sin \phi_i \\ \sin(\omega t - \phi_i) &= \sin \omega t \cos \phi_i - \cos \omega t \sin \phi_i\end{aligned}\quad (14.10g)$$

After much manipulation, the expressions for the contributions from right and left banks reduce to:

for the right bank:

$$\mathbf{F}_{s_R} \cong m_B r \omega^2 \left[\begin{aligned} &(\cos \omega t \cos \gamma - \sin \omega t \sin \gamma) \sum_{i=1}^{n/2} \cos \phi_i \\ &+ (\cos \omega t \sin \gamma + \sin \omega t \cos \gamma) \sum_{i=1}^{n/2} \sin \phi_i \\ &+ \frac{r}{l} (\cos 2\omega t \cos 2\gamma - \sin 2\omega t \sin 2\gamma) \sum_{i=1}^{n/2} \cos 2\phi_i \\ &+ \frac{r}{l} (\cos 2\omega t \sin 2\gamma + \sin 2\omega t \cos 2\gamma) \sum_{i=1}^{n/2} \sin 2\phi_i \end{aligned} \right] \hat{\mathbf{r}} \quad (14.10h)$$

and for the left bank:

$$\mathbf{F}_{s_L} \cong m_B r \omega^2 \begin{bmatrix} (\cos \omega t \cos \gamma + \sin \omega t \sin \gamma) \sum_{i=n/2+1}^n \cos \phi_i \\ -(\cos \omega t \sin \gamma - \sin \omega t \cos \gamma) \sum_{i=n/2+1}^n \sin \phi_i \\ + \frac{r}{l} (\cos 2\omega t \cos 2\gamma + \sin 2\omega t \sin 2\gamma) \sum_{i=n/2+1}^n \cos 2\phi_i \\ - \frac{r}{l} (\cos 2\omega t \sin 2\gamma - \sin 2\omega t \cos 2\gamma) \sum_{i=n/2+1}^n \sin 2\phi_i \end{bmatrix} \hat{\mathbf{i}} \quad (14.10i)$$

The summations in equations 14.10h and 14.10i give a set of sufficient criteria for **zero shaking force** through the second harmonic for each bank, similar to those for the inline engine in equations 14.3. We can resolve the shaking forces for each bank into components along and normal to the central X axis of the vee engine:*

$$\begin{aligned} F_{s_x} &= (F_{s_L} + F_{s_R}) \cos \gamma \hat{\mathbf{i}} \\ F_{s_y} &= (F_{s_L} - F_{s_R}) \sin \gamma \hat{\mathbf{j}} \\ \mathbf{F}_s &= F_{s_x} \hat{\mathbf{i}} + F_{s_y} \hat{\mathbf{j}} \end{aligned} \quad (14.10j)$$

* The effect of overbalancing the crank throws is not included in equation 14.10j as shown. The crankshaft is assumed to be exactly balanced here. See Appendix G for the complete equations that include the effects of crank overbalance. Program LINKAGES uses the equations from Appendix G to account for the effects of overbalancing in multicylinder engines.

Equations 14.10j provide additional opportunities for cancellation of shaking forces beyond the choice of phase angles; e.g., even with nonzero values of F_{s_L} and F_{s_R} , if γ is 90° , then the x component of the shaking force will be zero. Also, if $F_{s_L} = F_{s_R}$, then the y component of the shaking force will be zero for any γ . This situation obtains for the case of a horizontally opposed engine (see Section 14.8). With some vee or opposed engines it is possible to get cancellation of shaking force components even when the summations in equation 14.10 are not all zero.

The **shaking moment** equations are easily formed from the shaking force equations by multiplying each term in the summations by the moment arm as was done in equations 14.6. The moments exist within each bank, and their vectors will be orthogonal to the respective cylinder planes. For the right bank we define a moment unit vector $\hat{\mathbf{n}}$ perpendicular to the $\hat{\mathbf{r}}$ - Z plane in Figure 14-21. For the left bank we define a moment unit vector $\hat{\mathbf{m}}$ perpendicular to the $\hat{\mathbf{i}}$ - Z plane in Figure 14-21.

$$\mathbf{M}_{s_R} \cong m_B r \omega^2 \begin{bmatrix} (\cos \omega t \cos \gamma - \sin \omega t \sin \gamma) \sum_{i=1}^{n/2} z_i \cos \phi_i \\ + (\cos \omega t \sin \gamma + \sin \omega t \cos \gamma) \sum_{i=1}^{n/2} z_i \sin \phi_i \\ + \frac{r}{l} (\cos 2\omega t \cos 2\gamma - \sin 2\omega t \sin 2\gamma) \sum_{i=1}^{n/2} z_i \cos 2\phi_i \\ + \frac{r}{l} (\cos 2\omega t \sin 2\gamma + \sin 2\omega t \cos 2\gamma) \sum_{i=1}^{n/2} z_i \sin 2\phi_i \end{bmatrix} \hat{\mathbf{n}} \quad (14.11a)$$

$$\mathbf{M}_{s_L} \doteq m_B r \omega^2 \begin{bmatrix} (\cos \omega t \cos \gamma + \sin \omega t \sin \gamma) \sum_{i=n/2+1}^n z_i \cos \phi_i \\ -(\cos \omega t \sin \gamma - \sin \omega t \cos \gamma) \sum_{i=n/2+1}^n z_i \sin \phi_i \\ + \frac{r}{l} (\cos 2\omega t \cos 2\gamma + \sin 2\omega t \sin 2\gamma) \sum_{i=n/2+1}^n z_i \cos 2\phi_i \\ - \frac{r}{l} (\cos 2\omega t \sin 2\gamma - \sin 2\omega t \cos 2\gamma) \sum_{i=n/2+1}^n z_i \sin 2\phi_i \end{bmatrix} \hat{\mathbf{m}} \quad (14.11b)$$

The summations in equations 14.11a and 14.11b provide a set of sufficient criteria for **zero shaking moment** through the second harmonic for each bank, similar to those found for the inline engine in equations 14.7. Resolving the shaking moments for each bank into components along and normal to the central X axis of the vee engine gives:*

$$\begin{aligned} M_{s_x} &= (M_{s_L} - M_{s_R}) \sin \gamma \\ M_{s_y} &= (-M_{s_L} - M_{s_R}) \cos \gamma \\ \mathbf{M}_s &= M_{s_x} \hat{\mathbf{i}} + M_{s_y} \hat{\mathbf{j}} \end{aligned} \quad (14.11c)$$

Equation 14.11c allows possible cancellation of shaking moment components for some vee or opposed configurations even when the summations in equations 14.11a and 14.11b are not all zero; e.g., if γ is 90° , then the y component of the shaking moment is zero.

The **inertia torques** from the right and left banks of a vee engine are:

$$\mathbf{T}_{i21_R} \doteq \frac{1}{2} m_B r^2 \omega^2 \begin{bmatrix} \frac{r}{2l} \left(\sin(\omega t + \gamma) \sum_{i=1}^{n/2} \cos \phi_i - \cos(\omega t + \gamma) \sum_{i=1}^{n/2} \sin \phi_i \right) \\ - \left(\sin 2(\omega t + \gamma) \sum_{i=1}^{n/2} \cos 2\phi_i - \cos 2(\omega t + \gamma) \sum_{i=1}^{n/2} \sin 2\phi_i \right) \\ - \frac{3r}{2l} \left(\sin 3(\omega t + \gamma) \sum_{i=1}^{n/2} \cos 3\phi_i - \cos 3(\omega t + \gamma) \sum_{i=1}^{n/2} \sin 3\phi_i \right) \end{bmatrix} \hat{\mathbf{k}} \quad (14.12a)$$

$$\mathbf{T}_{i21_L} \doteq \frac{1}{2} m_B r^2 \omega^2 \begin{bmatrix} \frac{r}{2l} \left(\sin(\omega t - \gamma) \sum_{i=n/2+1}^n \cos \phi_i - \cos(\omega t - \gamma) \sum_{i=n/2+1}^n \sin \phi_i \right) \\ - \left(\sin 2(\omega t - \gamma) \sum_{i=n/2+1}^n \cos 2\phi_i - \cos 2(\omega t - \gamma) \sum_{i=n/2+1}^n \sin 2\phi_i \right) \\ - \frac{3r}{2l} \left(\sin 3(\omega t - \gamma) \sum_{i=n/2+1}^n \cos 3\phi_i - \cos 3(\omega t - \gamma) \sum_{i=n/2+1}^n \sin 3\phi_i \right) \end{bmatrix} \hat{\mathbf{k}} \quad (14.12b)$$

* The effect of overbalancing the crank throws is not included in equation 14.11c as shown. The crankshaft is assumed to be exactly balanced here. See Appendix G for the complete equations that include the effects of crank overbalance.

Program LINKAGES uses the equations from Appendix G to account for the effects of overbalancing in multicylinder engines.

Add the contributions from each bank for the total. For **zero inertia torque** through the third harmonic in a vee engine it is sufficient (but not necessary) that:

$$\begin{array}{cccc}
 \sum_{i=1}^{n/2} \sin \phi_i = 0 & \sum_{i=1}^{n/2} \cos \phi_i = 0 & \sum_{i=n/2+1}^n \sin \phi_i = 0 & \sum_{i=n/2+1}^n \cos \phi_i = 0 \\
 \sum_{i=1}^{n/2} \sin 2\phi_i = 0 & \sum_{i=1}^{n/2} \cos 2\phi_i = 0 & \sum_{i=n/2+1}^n \sin 2\phi_i = 0 & \sum_{i=n/2+1}^n \cos 2\phi_i = 0 \\
 \sum_{i=1}^{n/2} \sin 3\phi_i = 0 & \sum_{i=1}^{n/2} \cos 3\phi_i = 0 & \sum_{i=n/2+1}^n \sin 3\phi_i = 0 & \sum_{i=n/2+1}^n \cos 3\phi_i = 0
 \end{array} \quad (14.12c)$$

Note that when equations 14.12a and 14.12b are added, particular combinations of ϕ_i and γ may cancel the inertia torque even when some terms of equation 14.12c are nonzero.

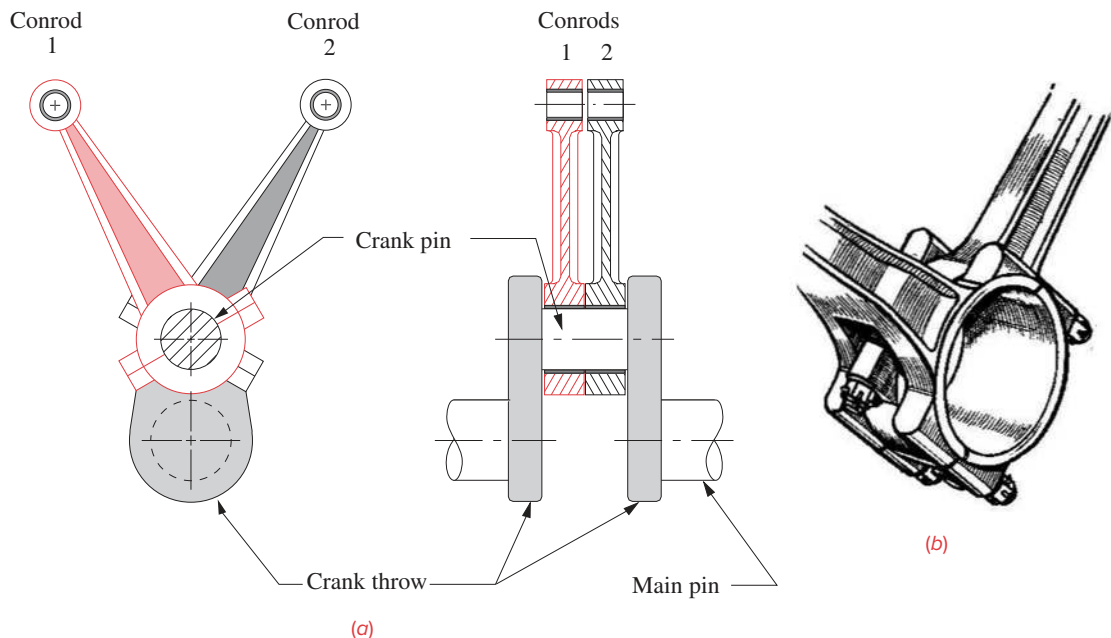
The **gas torque** is:

$$T_{g21} \cong F_g r \sum_{i=1}^n \left(\sin [\omega t - (\psi_i + \gamma_k)] \left\{ 1 + \frac{r}{l} \cos [\omega t - (\psi_i + \gamma_k)] \right\} \right) \hat{\mathbf{k}} \quad (14.13)$$

where the left bank has a *bank angle* $\gamma_k = +\gamma$ and the right bank a *bank angle* $\gamma_k = -\gamma$.

It is possible to design a vee engine that has as many crank throws as cylinders, but, for several reasons, this is not always done. The principal advantage of a vee engine over an inline of the same number of cylinders is its more compact size and greater stiffness. It can be about half the length of a comparable inline engine (at the expense of greater width), provided that the crankshaft is designed to accommodate two conrods per crank throw. The most common arrangement is to put the conrods side-by-side on a longer crankpin as shown in Figure 14-22a. Cylinders in opposite banks then share a crank throw, and one bank of cylinders is shifted along the crankshaft axis by the thickness of a conrod. A better balance condition is obtained by putting both conrods in the same plane using a fork-and-blade arrangement as in Figure 14-22b. In either case, the shorter, wider cylinder block and the shorter crankshaft are much stiffer in both torsion and bending than are those for an inline engine with the same number of cylinders. Figure 14-23 shows computer simulations of several bending and one torsional mode of vibration for a four throw crankshaft. The deflections are greatly exaggerated. The necessarily contorted shape of a crankshaft makes it difficult to control these deflections by design. If excessive in magnitude, they can lead to structural failure.

As an example we will now design the crankshaft for a four-stroke cycle, vee-eight engine. We could put two $\phi_i = 0, 180, 180, 0^\circ$ four-cylinder engines together on one such crankshaft and have the same balance conditions as the four-stroke, four-cylinder engine designed in the previous section (primaries balanced, secondaries unbalanced). However, the motivation for choosing that crankshaft for the four-cylinder engine was the need to space the four available power pulses evenly across the cycle. Equation 14.8b then dictated a 180° delta phase angle $\Delta\phi_i$ for that engine. Now we have eight cylinders available, and equation 14.8b defines a delta phase angle of 90° for optimum power pulse spacing. This means we could use the $\phi_i = 0, 90, 180, 270^\circ$ crankshaft designed for the two-stroke

**FIGURE 14-22**

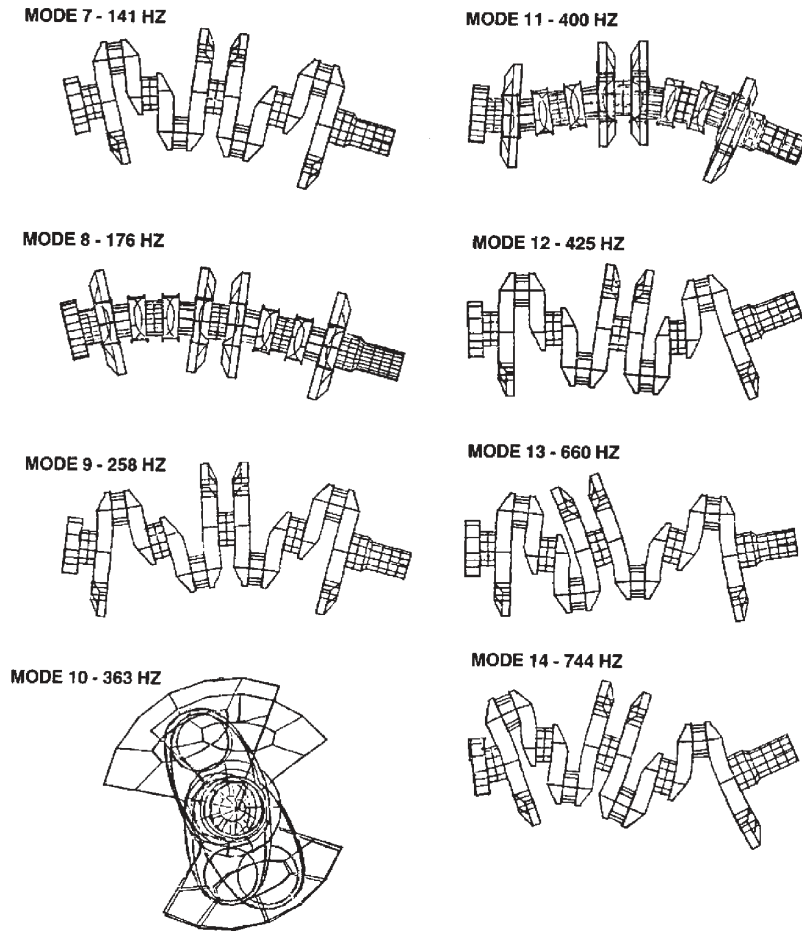
Two connecting rods on a common crank throw (a) side-by-side, (b) fork-and-blade *Public domain*

four-cylinder engine shown in Figure 14-12 and take advantage of its better inertia balance condition as well as achieve even firing in the four-stroke eight-cylinder vee engine.

The $\phi_i = 0, 90, 180, 270^\circ$ four-cylinder crankshaft has all inertia factors equal to zero except for the primary and secondary moments. We learned that arranging the crank throws with mirror symmetry about the midplane would balance the primary moment. Some thought and/or sketches will reveal that it is not possible to obtain this mirror symmetry with any four-throw, 90° delta phase angle crankshaft arrangement. However, just as rearranging the order of the crank throws from $\phi_i = 0, 180, 0, 180^\circ$ to $\phi_i = 0, 180, 180, 0^\circ$ had an effect on the shaking moments, rearranging this crankshaft's throw order will as well. A crankshaft of $\phi_i = 0, 90, 270, 180^\circ$ has all inertia factors equal to zero except for the primary moment. The secondary moment is now gone.* This is an advantage worth taking. We will use this crankshaft for the vee-eight and deal with the primary moment later.

Figure 14-24a shows the crank phase diagram for the right bank of a vee-eight engine with a $\phi_i = 0, 90, 270, 180^\circ$ crankshaft. Figure 14-24b shows the crank phase diagram for the second (left) bank which is identical to that of the right bank (as it must be since they share crank throws), but it is *shifted to the right by the vee angle 2γ* . Note that in Figure 14-21, the two pistons are driven by conrods on a common crank throw with positive ω , and the piston in the right bank will reach TDC before the one in the left bank. Thus as we show it, the left bank's piston motions lag those of the right bank. Lagging events occur later in time, so we must shift the second (left) bank rightward by the vee angle on the crank phase diagram.

* The explanation for this is quite simple. Equation 14.7b shows that second moments are a function of twice the phase angles and the cylinder moment arms. If you double the values of the original $0, 90, 180, 270^\circ$ phase angle sequence and modulo them with 360 , you get $0, 180, 0, 180^\circ$ which is not mirror-symmetric. Doubling the new phase angle sequence of $0, 90, 270, 180^\circ$, modulo 360 , gives $0, 180, 180, 0^\circ$ which is mirror-symmetric. It is this symmetry of the doubled phase angles that causes cancellation of the second harmonic of the shaking moment.

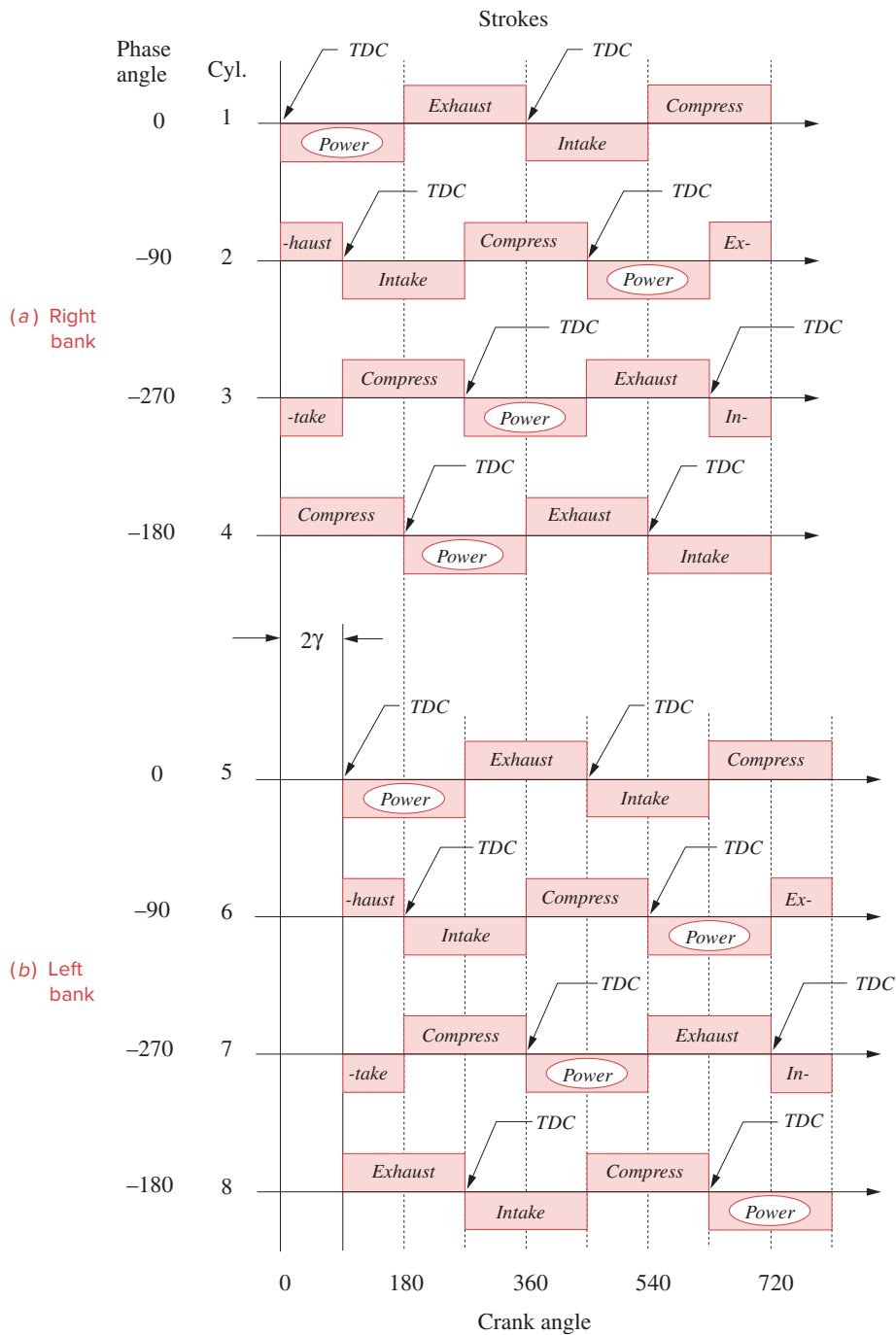
**FIGURE 14-23**

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Bending and torsional modes of vibration in a four-throw crankshaft

We would like to shift the second bank of cylinders such that its power pulses are evenly spaced among those of the first bank. A little thought (and reference to equation 14.8b) should reveal that, in this example, each four-cylinder bank has potentially $720/4 = 180^\circ$ between power pulses. Our chosen crank throws are spaced at 90° increments. A 90° vee angle (bank angle $\gamma = 45^\circ$) will be optimum in this case as the phase angles and bank angles will add to create an effective spacing of 180° . Every vee-engine design of four or more cylinders will have one or more optimum vee angles that will give approximately even firing with any particular set of crank phase angles.

Several firing orders are possible with this many cylinders. Vee engines are often arranged to fire cylinders in opposite banks successively to balance the fluid flow demands in the intake manifold. Our cylinders are numbered from front to back, first down the right

**FIGURE 14-24**

Four-stroke vee-eight crank phase diagram with 0, 90, 270, 180° crankshaft phase angles

90 deg Vee, 8-Cylinder
4-Stroke Cycle

Bore = 2.50 in

Stroke = 2.55

B/S = 0.98

L/R = 3.50

$m_A = 0.0418$ bl

$m_B = 0.0116$ bl

RPM = 3400

Phase Angles:

0 90 270 180 0 90 270 180

Power Strokes:

0 90 180 270 360 450 540 630

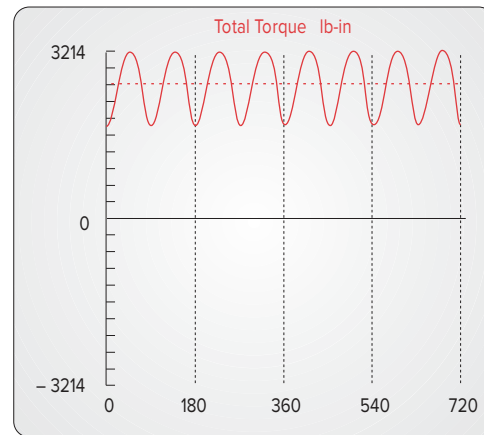


FIGURE 14-25

Total torque in the 90° vee-eight engine with 0, 90, 270, 180° crankshaft phase angles

bank and then down the left bank. The firing order shown in Figure 14-24b is 1, 5, 4, 3, 7, 2, 6, 8 and results in power stroke angles $\psi_i = 0, 90, 180, 270, 360, 450, 540, 630^\circ$. This will clearly give even firing with a power pulse every 90° .

Figure 14-25 shows the total torque for this engine design, which in this case is equal to the gas torque because the inertia torque is zero. Table 14-5 and Figure 14-26 show the only significant unbalanced inertial component in this engine to be the primary moment, which is quite large. The fourth harmonic terms have negligible coefficients in the Fourier series, and we have truncated them from the equations. We will address the balancing of this primary moment in a later section of this chapter.

Any vee-cylinder configuration may have one or more desirable vee angles which will give both even firing and acceptable inertia balance. However, vee engines of fewer than

TABLE 14-5 Force and Moment Balance State of an 8-Cylinder, Vee Engine with a 0, 90, 270, 180° Crankshaft, and $z_1 = 0, z_2 = 1, z_3 = 2, z_4 = 3$

Primary forces in each bank:	$\sum_{i=1}^n \sin \phi_i = 0$	$\sum_{i=1}^n \cos \phi_i = 0$
Secondary forces in each bank:	$\sum_{i=1}^n \sin 2\phi_i = 0$	$\sum_{i=1}^n \cos 2\phi_i = 0$
Primary moments in each bank:	$\sum_{i=1}^n z_i \sin \phi_i = -1$	$\sum_{i=1}^n z_i \cos \phi_i = -3$
Secondary moments in each bank:	$\sum_{i=1}^n z_i \sin 2\phi_i = 0$	$\sum_{i=1}^n z_i \cos 2\phi_i = 0$

90 deg Vee, 8-Cylinder
4-Stroke Cycle

Bore = 2.50 in

Stroke = 2.55

B/S = 0.98

L/R = 3.50

m_A = 0.0418 lb

m_B = 0.0116 lb

RPM = 3400

Phase Angles:

0 90 270 180 0 90 270 180

Power Strokes:

0 90 180 270 360 450 540 630

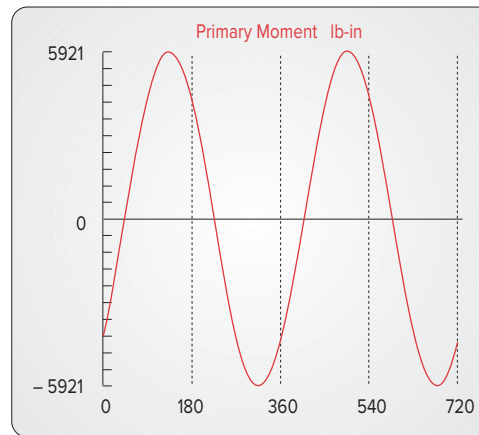


FIGURE 14-26

Unbalanced primary moment in the 90° vee-eight engine with a 0, 90, 270, 180° crankshaft

twelve cylinders will not be completely balanced by means of their crankshaft configuration. The desirable vee angles will typically be an integer multiple (including one) or submultiple of the optimum delta phase angle as defined in equations 14.8 for that engine. Ninety degrees is the optimum vee angle (2γ) for an eight-cylinder vee engine. To see the results for this vee-eight engine configuration, run program LINKAGES and select the vee-eight from the *Examples* pulldown menu. See Appendix A for instructions on the use of the program.

14.8 OPPOSED ENGINE CONFIGURATIONS

An opposed engine is essentially a vee engine with a 180° vee angle. The advantage, particularly with a small number of cylinders such as two or four, is the relatively good balance condition possible. A four-stroke opposed (boxer) twin* with 0, 180° crank has even firing plus primary and secondary force balance, though all harmonics of its moment are nonzero. A four-stroke, opposed four-cylinder engine (flat four) with a 0, 180, 180, 0°, four-throw crank is even firing and has primary force and moment balance. Unlike its inline-four cousin with the same crankshaft, it also has secondary force balance from equation 14.10j. A four-stroke opposed six with a six-throw, 0, 120, 240, 60, 180, 300° crank has even firing and the same good balance condition as the inline six. Program LINKAGES will calculate the parameters for opposed as well as vee and inline configurations.

14.9 BALANCING MULTICYLINDER ENGINES *Watch a Video on Balancing Multicylinder Engines (31:12)*[§]

With a sufficient number (m) of cylinders, properly arranged in banks of n cylinders in a multibank engine,[†] an engine can be inherently balanced. In a two-stroke engine with its crank throws arranged for even firing, all harmonics of shaking force will be balanced except those whose harmonic number is a multiple of n . In a four-stroke inline engine

* As in the BMW R-series motorcycles.

[§] http://www.designofmachinery.com/DOM/Balancing_Multicylinders.mp4

[†] For an inline engine, $m = n$.

with its crank throws arranged for even firing, all harmonics of shaking force will be balanced except those whose harmonic number is a multiple of $n/2$. Primary shaking moment components will be balanced if the crankshaft is mirror-symmetric about the central transverse plane. A four-stroke inline configuration then requires at least six cylinders to be inherently balanced through the second harmonic. We have seen that an inline four with a 0, 180, 180, 0° crankshaft has nonzero secondary forces and moments as well as nonzero inertia torque. The inline six with a mirror-symmetric crank of $\phi_i = 0, 240, 120, 120, 240, 0^\circ$ will have zero shaking forces and moments through the second harmonic, though the inertia torque's third harmonic will still be present. To see the results of this six-cylinder inline engine configuration, run program LINKAGES and select the inline six from the *Examples* pulldown menu.

A VEE-TWELVE is then the smallest vee engine with an inherent state of near perfect balance, as it is two inline sixes on a common crankshaft. We have seen that vee engines generally take on the balance characteristics of the inline banks from which they are made. Equations 14.10a to 14.10i and 14.11a and 14.11b introduced no new criteria for balance in the vee engine over those already defined in equations 14.3 and 14.5 for shaking force and moment balance in the inline engine except for the possible cancellation available from the value of bank angle γ in equations 14.10j and 14.11c.* Open the file BMWV12.eng in program LINKAGES to see the results for a vee-twelve engine. The common vee-eight engine with crankshaft phase angles of $\phi_i = 0, 90, 270, 180^\circ$ has an unbalanced primary moment as does the inline four from which it is made. It is an example in program LINKAGES.

UNBALANCED INERTIA TORQUES can be smoothed with a flywheel as was shown in Section 13.8 for the single-cylinder engine. Note that even an engine having zero inertial torque may require a flywheel to smooth its variations in gas torque. The total torque function should be used to determine the energy variations to be absorbed by a flywheel as it contains both gas torque and inertia torque (if any). The method of Section 11.11 also applies to calculation of the flywheel size needed in an engine, based on its variation in the total torque function. Program LINKAGES will compute the areas under the total torque pulses needed for the calculation. See the referenced sections for the proper flywheel design procedure.

UNBALANCED SHAKING FORCES AND SHAKING MOMENTS can be cancelled by the addition of one or more rotating balance shafts within the engine. To cancel the primary components usually requires two balance shafts rotating at crank speed, one of which can be the crankshaft itself. To cancel the secondary components usually requires at least two balance shafts rotating at twice crank speed, gear or chain driven from the crankshaft. Figure 14-27a shows a pair of counterrotating shafts which are fitted with eccentric masses arranged 180° out of phase.† As shown, the unbalanced centrifugal forces from the equal, unbalanced masses will add to give a shaking force component in the vertical direction of twice the unbalanced force from each mass, while their horizontal components will exactly cancel. Pairs of counterrotating eccentrics can be arranged to provide a harmonically varying force in any one plane. The harmonic frequency will be determined by the rotational speed of the shafts.

If we arrange two pairs of eccentrics, with one pair displaced some distance along the shaft from the other, and also rotated 180° around the shaft from the first, as shown

* The angle γ is most effective when set to 90° for an opposed engine with $2\gamma = 180^\circ$ since its cosine is zero, which cancels some components of the shaking force and moment.

† This is called a Lanchester balancer after its English inventor who developed it prior to World War I (c. 1913). It is still used in various kinds of machinery as well as in engines to cancel inertia forces.

in Figure 14-27b, we will get a harmonically varying couple in one plane. There will be cancellation of the forces in one direction and summation in an orthogonal direction.

Thus to cancel the shaking moment in any plane, we can arrange a pair of shafts, each containing two eccentric masses displaced along those shafts, 180° out of phase, and gear the shafts together to rotate in opposite directions at any multiple of crankshaft speed. To cancel the shaking force as well, it is only necessary to provide sufficient additional unbalanced mass in one of the pairs of eccentric masses to generate a shaking force opposite to that of the engine, over and above that needed to generate the forces of the couple.

In an inline or opposed engine, the unbalanced forces and moments are all confined to the single plane of the cylinders as they are due entirely to the reciprocating masses assumed concentrated at the wrist pin. (We are assuming that all crank throws are exactly balanced rotationally to cancel the effects of mass at the crank pin.) In a vee engine, however, the shaking forces and moments have both x and y components as shown in equations 14.10 and 14.11 and in Figure 14-21. The shaking effects of each bank's pistons are acting within the plane of that bank's cylinders, and the bank angle γ is used to resolve them into x and y components using equations 14.10j and 14.11c.

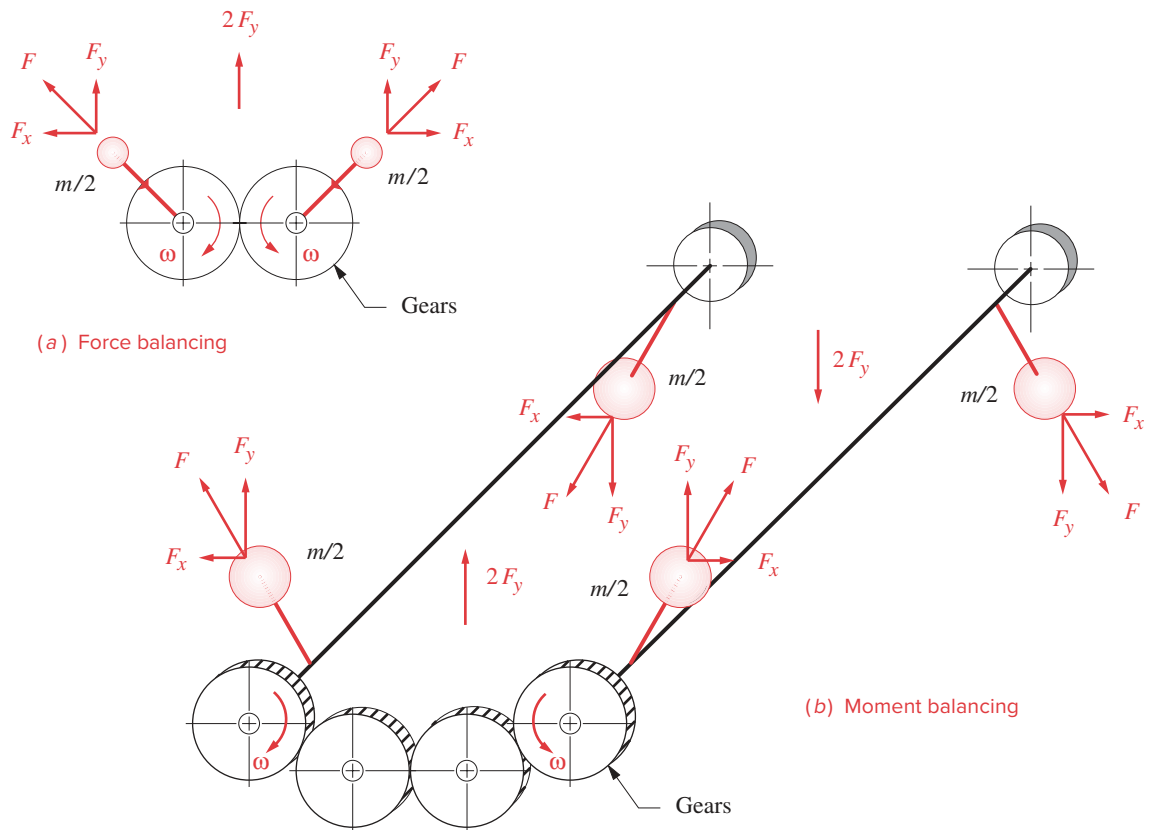


FIGURE 14-27

Counterrotating eccentric masses can balance forces and moments.

90 deg Vee, 2-Cylinder
4-Stroke Cycle

Bore = 3.20 in

Stroke = 3.11

B/S = 1.03

L/R = 3.50

$m_A = 0.0418$ lb

$m_B = 0.0116$ lb

RPM = 3400

Phase Angles:

0 0

Power Strokes:

0 450

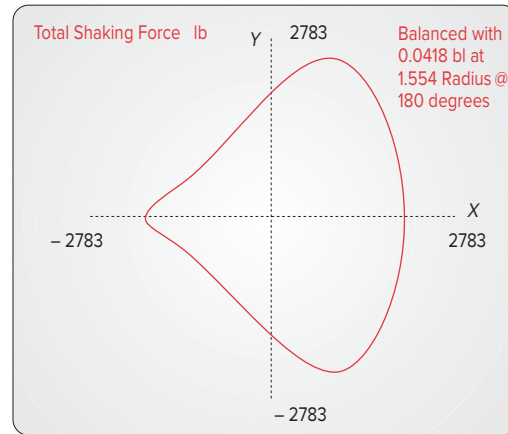


FIGURE 14-28

Shaking force in a 90° vee-twin engine (looking end-on to the crankshaft axis)

VEE-TWIN Primary force balance is possible in a 4-stroke vee-twin of any vee angle v if two crank pins are used. If $\phi_1 = 0$, the phase angle of the second pin ϕ_2 must be:^[2]

$$\phi_2 = 180^\circ - 2v \quad (14.14a)$$

For even firing, the relationship must be:

$$\phi_2 = 360^\circ - v \quad (14.14b)$$

where v is the vee angle as defined in Figure 14-21. The only value of v that satisfies both criteria is 180° (opposed cylinders). All other vee-twin angles can have either even firing or primary balance, but not both.*

Figure 14-28 shows the two-dimensional shaking force present in a two-cylinder 90° vee, single-crank-pin engine, which satisfies equation 14.14a with $\phi_1 = \phi_2 = 0$, coalescing its “two crank pins” into one for this vee angle. The inertia force of each piston is confined to the reciprocating plane (bank) of that piston, but the vee angle between the cylinder banks creates the pattern shown when the primary and secondary components of each piston force are added vectorially. The shaking force of the 90° vee, single-crank pin twin has a rotating primary component of constant magnitude that can be cancelled with overbalanced counterweights on the crankshaft. However, its second harmonic is planar (in the YZ plane). To cancel it requires a pair of twice-speed balance shafts as shown in Figure 14-27a.

VEE EIGHT The 90° vee-eight engine with a 0, 90, 270, 180° crankshaft, which has only an unbalanced primary moment, presents a special case. The 90° angle between the banks results in equal horizontal and vertical components of the primary shaking moment that reduces it to a couple of constant magnitude rotating about the crank axis at crankshaft speed in the same direction as the crank as shown in Figure 14-29. With this vee-eight engine, the primary moment can be balanced by merely adding two eccentric counterweights of proper size and opposite orientation to the crankshaft. No independent,

* Vee-twins for motor-cycles have been made in a variety of vee angles: 45°, 48°, 50°, 52°, 60°, 75°, 80°, 90°, 180°, and possibly others. All but the 180° boxer twin have unbalanced secondary force and most have unbalanced primary force. Some have been fitted with balance shafts to reduce shaking. Many are also uneven firing, giving them a distinctive exhaust sound. The Harley Davidson single-crankpin, 45° vee-twin is one example whose sound (which the company unsuccessfully tried to patent) has been described as “potato-potato.”

90 deg Vee 8 Cylinder
4 Stroke Cycle

Bore = 2.50 in

Stroke = 2.55

B/S = 0.98

L/R = 3.50

$m_A = 0.0418$ bl

$m_B = 0.0116$ bl

RPM = 3400

Phase Angles:

0 90 270 180 0 90 270 180

Power Strokes:

0 90 180 270 360 450 540 630

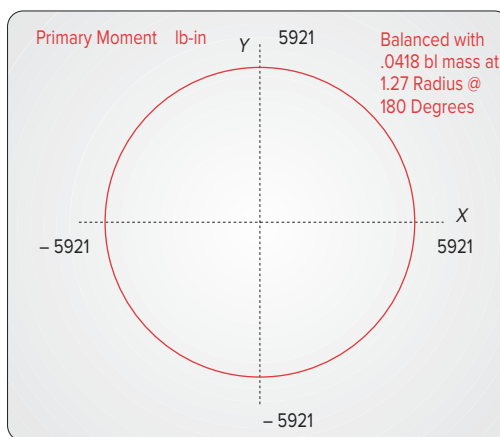


FIGURE 14-29

Primary moment in the 90° vee-eight engine (looking end-on to the crankshaft axis)

second balance shaft is needed in the 90° vee-eight engine with this crankshaft. The 180° out-of-phase counterweights are typically placed near the ends of the crankshaft to obtain the largest moment arm possible and thus reduce their size.

VEE-SIX engines with a 3-throw, 2-conrod-per-throw, 0, 240, 120° crankshaft have unbalanced primary and secondary moments as does the three-cylinder inline from which they are made and are not even firing. A 3-throw, 2-conrod-per-throw vee six needs a 120° vee angle for inherent balance. To reduce engine width, vee-sixes are most often made with a 60° vee angle which gives even firing with a 6-throw 0, 240, 120, 60, 300, 180° crank. This engine has unbalanced primary and secondary moments, each being a constant-magnitude rotating vector like that of the vee-eight shown in Figure 14-29. The primary component can be completely balanced by adding counterweights to the crankshaft as done in the 90° vee-eight. Some manufacturers also add a single balance shaft in the valley of the-vee six, driven by gears at twice crankshaft speed to cancel the circular, constant-magnitude secondary shaking moment. Some vee-sixes use 90° vee angles to allow assembly on the same production line as 90° vee eights, but 3-throw 90° vee-sixes will run roughly due to uneven firing unless the crankshaft is redesigned to shift (or splay) the two conrods on each pin by 30°. This results in a 4-main bearing, 6-throw 0, 240, 120, 30, 270, 150° crankshaft that gives even firing but has nonconstant-magnitude primary and secondary shaking moments.

Calculation of the magnitude and location of the eccentric balance masses needed to cancel any shaking forces or moments is a straightforward exercise in **static balancing** (for forces) and **two-plane dynamic balancing** (for moments) as discussed in Sections 12.1 and 12.2, respectively. The unbalanced forces and moments for the particular engine configuration are calculated from the appropriate equations in this chapter. Two correction planes must be selected along the length of the balance shafts/crankshaft being designed. The magnitude and angular locations of the balance masses can then be calculated by the methods described in the noted sections of Chapter 12.

Secondary Harmonic Balancing of the Four-Cylinder Inline Engine

The four-cylinder inline engine with a 0, 180, 180, 0° crankshaft is one of the most widely used engines in the automobile industry. As described in a previous section, this engine suffers from unbalanced secondary force, moment, and torque. If the displacement of the engine is less than about 2.0 liters, then the magnitudes of the secondary forces may be small enough to be ignored, especially if the engine mounts provide good vibration isolation of the engine from the passenger compartment. Above that displacement, objectionable noise, vibration, and harshness (NVH) may be heard and felt by the passengers at certain engine speeds where the frequency of the engine's second harmonic coincides with one of the natural frequencies of the body structure. Then some balancing is needed in the engine to avoid customer dissatisfaction.

Equation 14.2d defines the inline engine shaking force. Applying the relevant factors from Table 14-4 for this engine to the second harmonic term gives

$$\mathbf{F}_{s_2} \cong m_B r \omega^2 \frac{4r}{l} \cos 2\omega t \hat{\mathbf{i}} \quad (14.15)$$

The shaking torque for an inline engine is given by equation 14.4c in combination with equation 13.15f. Taking only the second harmonic term and applying the relevant factors from Table 14-4 for this engine give

$$\mathbf{T}_{s_2} \cong 2m_B r^2 \omega^2 \sin 2\omega t \hat{\mathbf{k}} \quad (14.16)$$

The principle of the Lanchester balancer, shown in Figure 14-27a, can be used to counteract the secondary forces by driving its two counterrotating balance shafts at twice crankshaft speed with chains and/or gears. Figure 14-30 shows such an arrangement as applied to a Mitsubishi 2.6-liter, four-cylinder engine.*

H. Nakamura^[3] improved on Lanchester's 1913 design by arranging the balance shafts within the engine so as to cancel the second harmonic of the inertia torque as well as the secondary inertia force. But, his arrangement does not affect the unbalanced secondary shaking moment. In fact, it is designed to impart zero net moment about a transverse axis to either balance shaft in order to minimize bending moments on the shafts, and so reduce bearing loads and friction losses. This feature is the subject and principal claim of Nakamura's patent on this design.^[4]

Figure 14-31a shows a schematic of a conventional Lanchester balancer arranged with the two counterrotating balance shafts with their centers in a single horizontal plane transverse to the vertical plane of piston motion.[†]

The balance force from the two balance shafts combined is

$$\mathbf{F}_{bal} \cong -8m_{bal}r_{bal}\omega^2 \cos 2\omega t \hat{\mathbf{i}} \quad (14.17)$$

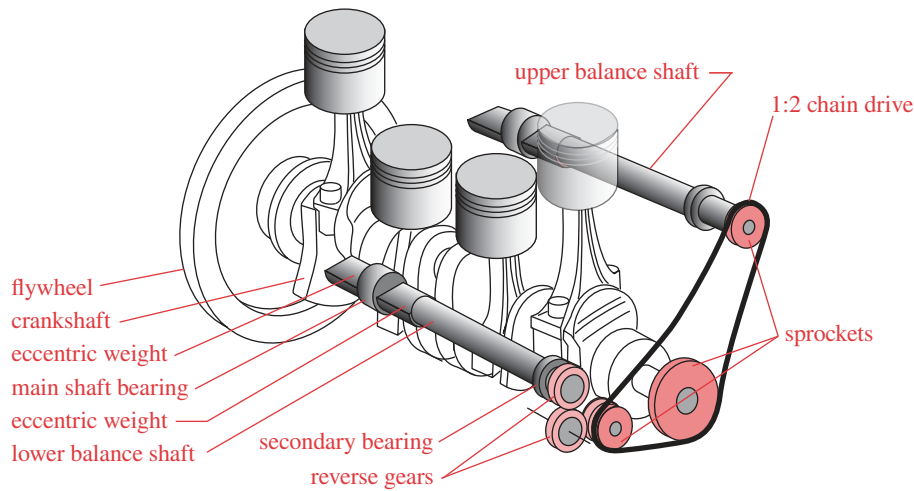
where m_{bal} and r_{bal} are the mass and radius, respectively, of one balance weight.

Figure 14-31b shows Nakamura's arrangement of the balance shafts with one situated above the other in separate horizontal planes. The vertical offset $x_1 - x_2$ between the shafts, in combination with the oppositely directed but equal-magnitude horizontal components of the counterweights' centrifugal forces, creates a time-varying couple *about the crankshaft axis* defined as:

$$\mathbf{T}_{bal} \cong -4m_{bal}r_{bal}\omega^2 (x_1 - x_2) \sin 2\omega t$$

* Also used by Chrysler and Porsche (in the 244) under license from Mitsubishi.

† The arrangement shown in Figure 14-31a is only schematic of Lanchester's original design in which the crankshaft drove the balance shafts through right-angled helical gears with the balance shaft axes parallel to the y axis of Figure 14-31, i.e., transverse to, rather than parallel to, the crankshaft axis as shown here. See reference [4] for drawings of his original design.

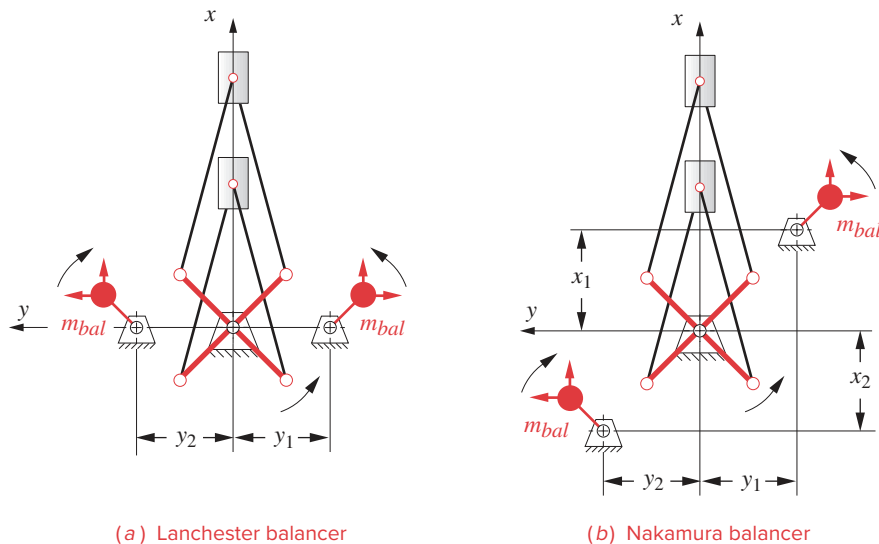
**FIGURE 14-30**

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Balance shafts used to eliminate the secondary unbalance in the four-cylinder inline engine

where x and y refer to the coordinates of the shaft centers referenced to the crankshaft center, and the subscripts 1 and 2 refer, respectively, to the balance shaft turning in directions the same as and opposite to that of the crankshaft.

The vertical components of the balance weights' centrifugal forces still add to provide force balance as in equation 14.17. The torque in equation 14.18 will have opposite sense to the shaking torque if the upper shaft turns in the same direction as, and the lower shaft turns in the opposite direction to that of the crankshaft.



(a) Lanchester balancer

(b) Nakamura balancer

FIGURE 14-31

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Two types of secondary balancer mechanisms for the four-cylinder inline engine

For force balance, equations 14.15 and 14.17 must sum to zero,

$$m_B r \omega^2 \frac{4r}{l} \cos 2\omega t - 8m_{bal} r_{bal} \omega^2 \cos 2\omega t = 0$$

$$\text{or} \quad m_{bal} r_{bal} = \frac{r}{2l} m_B r \quad (14.19)$$

which defines the mass-radius product needed for the balance mechanism.

For torque balance, equations 14.16 and 14.18 must sum to zero.

$$2m_B r^2 \omega^2 \sin 2\omega t - 4m_{bal} r_{bal} \omega^2 (x_1 - x_2) \sin 2\omega t - (y_1 + y_2) \cos 2\omega t = 0 \quad (14.20a)$$

Substitute equation 14.19 in 14.20a.

$$2m_B r^2 \sin 2\omega t - 4 \frac{r}{2l} m_B r (x_1 - x_2) \sin 2\omega t - (y_1 + y_2) \cos 2\omega t = 0 \quad (14.20b)$$

For this equation to be zero for all ωt ,

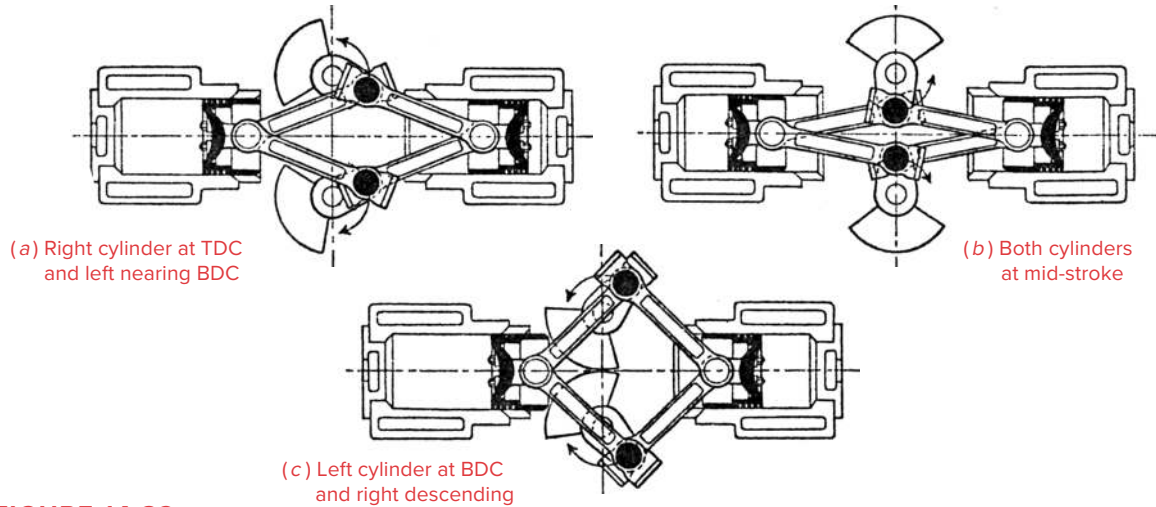
$$\begin{aligned} y_2 &= -y_1 \\ x_1 - x_2 &= l \end{aligned} \quad (14.20c)$$

So, if the balance shafts are arranged symmetrically with respect to the piston plane at any convenient locations y_1 and $-y_1$, and the distance $x_1 - x_2$ is made equal to the length of the connecting rod l , then the second harmonic of the inertia torque will be completely cancelled. Since the second harmonic is the only nonzero component of inertia torque in this engine as can be seen in Figure 14-20b, it will now be completely balanced for shaking force and shaking torque (but not shaking moment).

There is also significant oscillation of the gas torque in a four-cylinder engine as shown in Figure 14-20a. The gas torque is 180° out of phase with the inertia torque as can be seen in Figure 14-20b and so provides some natural cancellation as shown in the total torque curve of Figure 14-20c. The magnitude of the gas torque varies with engine load and so cannot itself be cancelled with any particular balance shaft geometry for all conditions. However, one engine speed and load condition can be selected as representative of the majority of typical driving conditions, and the balance system geometry altered to give an optimum reduction of total engine torque under those conditions. Nakamura estimates that gas torque magnitude is about 30% of the inertia torque under typical driving conditions and so suggests a value of $x_1 - x_2 = 0.7l$ for the best overall reduction of total torque oscillation in this engine. Note that the average value of the driving torque is not affected by balancing because the average torque of any rotating balance system is always zero.

A Perfectly Balanced Two-Cylinder Engine

Frederick Lanchester, in 1897, devised an extremely clever horizontally opposed engine arrangement^[5] of Figure 14-32 that, with only two cylinders, completely cancelled all harmonics of inertia forces and moments. He recognized that the lateral motion of the connecting rods was a contributor and so provided two counterrotating crankshafts, driven by a total of six connecting rods, three per crank pin, with two upper rods straddling one lower rod for Z axis symmetry. The crank counterweights exactly balance the cranks. The **colinear** opposed pistons exactly balance one another's linear accelerations and the

**FIGURE 14-32**

Perfectly balanced Lanchester two-cylinder, horizontally opposed engine (1897) *Public Domain*

scissors action of the multiple conrods exactly cancels all higher harmonics of motion. Clearly the work of genius. The genesis of his later (c. 1913) harmonic balancer of Figure 14-31a can also be seen here.

14.10 REFERENCES

- 1 **MacKay, R. F.** (1915). *The Theory of Machines*, Edward Arnold: London, p. 103.
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- 3 **Nakamura, H.** (1976). "A Low Vibration Engine with Unique Counter-Balance Shafts." SAE Paper: 760111.
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- 6 **Ibid.**, p. 26.

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TABLE P14-0

Topic/Problem Matrix

14.5 Shaking Moment in Inline Engines

14-8, 14-9

14.6 Even Firing14-1, 14-2, 14-3,
14-4, 14-5, 14-6,
14-7, 14-19, 14-20**14.7 Vee Engine Configurations**14-10, 14-11, 14-12,
14-20, 14-21**14.8 Opposed Engine Configurations**

14-13, 14-14

14.9 Balancing Multicylinder Engines14-15, 14-16, 14-17,
14-18, 14-23, 14-24

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

* More information on this engine design can be found in program LINKAGES where it is one of the examples.

14.12 PROBLEMS

- 14-1 Draw a crank phase diagram for a three-cylinder inline engine with a 0, 120, 240° crankshaft and determine all possible firing orders and select the best arrangement to give even firing for each stroke cycle for:
- Four-stroke cycle
 - Two-stroke cycle
- 14-2 Repeat Problem 14-1 for an inline four-cylinder engine with 0, 90, 270, 180° crank.
- 14-3 Repeat Problem 14-1 for a 45° vee, four-cylinder engine with 0, 90, 270, 180° crank.
- 14-4 Repeat Problem 14-1 for a 45° vee, two-cylinder engine with 0, 90° crank.
- 14-5 Repeat Problem 14-1 for a 90° vee, two-cylinder engine with 0, 180° crank.
- 14-6 Repeat Problem 14-1 for a 180° opposed, two-cylinder engine with a 0, 180° crank.
- 14-7 Repeat Problem 14-1 for a 180° opposed, four-cylinder engine with 0, 180, 180, 0° crank.
- †14-8 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-1.
- †14-9 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-2.
- †14-10 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-3.
- †14-11 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-4.
- †14-12 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-5.
- †14-13 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-6.
- †14-14 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-7.
- 14-15 Derive expressions, in general terms, for the magnitude and angle with respect to the first crank throw of the mass-radius products needed on the crankshaft to balance the shaking moment in a 90° vee-eight engine with a 0, 90, 270, 180° crankshaft.
- 14-16 Repeat Problem 14-15 for a 90° vee-six with a 0, 240, 120° crankshaft.
- 14-17 Repeat Problem 14-15 for a 90° vee-four with a 0, 180° crankshaft.
- †14-18 Design a pair of Nakamura balance shafts to cancel the shaking force and reduce torque oscillations in the engine shown in Figure 14-19.*
- 14-19 Using program LINKAGES, data in Table P14-1, and the crank phase diagram from Problem 14-1, determine the maximum force magnitudes on main pin, crank pin, wrist pin, and piston for a 2-stroke engine with even firing. Overbalance the crank, if necessary, to bring the shaking force down to at least half the unbalanced value.
- 14-20 Using program LINKAGES, data in Table P14-1, and the crank phase diagram from Problem 14-2, determine the maximum force magnitudes on main pin, crank pin, wrist pin, and piston for a 4-stroke engine with even firing. Overbalance the crank, if necessary, to bring the shaking force down to at least half of the unbalanced value.

- 14-21 Using program LINKAGES, data in Table P14-1, and the crank phase diagram from Problem 14-3, determine the maximum force magnitudes on main pin, crank pin, wrist pin, and piston for a 4-stroke engine with even firing. Overbalance the crank, if necessary, to bring the shaking force down to at least half of the unbalanced value.
- 14-22 Using program LINKAGES, data in Table P14-1, and the crank phase diagram from Problem 14-4, determine the maximum force magnitudes on main pin, crank pin, wrist pin, and piston for a 2-stroke engine with even firing. Overbalance the crank, if necessary, to bring the shaking force down to at least half of the unbalanced value.
- ^{†*} 14-23 A four-cylinder inline engine with a 0, 180, 180, 0° crankshaft has a stroke of $S = 3.50$ in, a conrod length to crank radius ratio of $L/R = 3.75$, and an effective wrist pin mass of $m_B = 0.0215$ blob. Design a pair of Nakamura balance shafts to cancel the shaking force and reduce the oscillations in the engine.
- [†] 14-24 Repeat problem 14-23 with $S = 2.750$ in, $L/R = 3.00$, and $m_B = 0.0125$ blob.

14.13 PROJECTS

These are loosely structured design problems intended for solution using program LINKAGES. All involve the design of one or more multicylinder engines and differ mainly in the specific data for the engine. The general problem statement is:

Design a multicylinder engine for a specified displacement and stroke cycle. Optimize the conrod/crank ratio and bore/stroke ratio to minimize shaking forces, shaking torque, and pin forces, also considering package size. Design your link shapes and calculate realistic dynamic parameters (mass, CG location, moment of inertia) for those links using the methods of Chapters 10-13. Dynamically model the links as described in those chapters. Balance or overbalance the linkage as needed to achieve the desired results. Choose crankshaft phase angles (and vee angles, if appropriate) to optimize the inertial balance of the engine. Choose a firing order and determine the power stroke angles to optimize even firing. Trade off inertia balance if necessary to achieve even firing. Design and size a minimum-weight flywheel by the method of Chapter 11 to smooth total torque. Write an engineering report on your design and analysis.

- P14-1 Two-stroke cycle inline twin with a displacement of 1 liter.
- P14-2 Four-stroke cycle inline twin with a displacement of 1 liter.
- P14-3 Two-stroke cycle vee-twin with a displacement of 1 liter.
- P14-4 Four-stroke cycle vee-twin with a displacement of 1 liter.
- P14-5 Two-stroke cycle opposed twin with a displacement of 1 liter.
- P14-6 Four-stroke cycle opposed twin with a displacement of 1 liter.
- P14-7 Two-stroke cycle vee-four with a displacement of 2 liters.
- P14-8 Four-stroke cycle vee-four with a displacement of 2 liters.
- P14-9 Two-stroke cycle opposed four with a displacement of 2 liters.
- P14-10 Four-stroke cycle opposed four with a displacement of 2 liters.
- P14-11 Two-stroke inline five-cylinder with a displacement of 2.5 liters.
- P14-12 Four-stroke inline five-cylinder with a displacement of 2.5 liters.
- P14-13 Two-stroke cycle vee-six with a displacement of 3 liters.

TABLE P14-1

Data for Problems 14-19 to 14-22

Displacement	10.0
Bore	1.87
L/R ratio	3.00
r_{G2}/r	0.40
r_{G3}/l	0.36
Main pin dia	2.00
Crank pin dia	1.50
Idle rpm	600
Redline rpm	4000
Piston mass	0.015
Conrod mass	0.012
Crank mass	0.045
P_{gmax}	5500
Friction coeff	0.02
Flywheel coeff	0.10

* Answers in Appendix F.

[†] These problems are suited to solution using *Mathcad*, *Matlab*, or, *TKSolver* equation solver programs.

- P14-14 Four-stroke cycle vee-six with a displacement of 3 liters.
- P14-15 Two-stroke cycle opposed six with a displacement of 3 liters.
- P14-16 Four-stroke cycle opposed six with a displacement of 3 liters.
- P14-17 Two-stroke inline seven-cylinder with a displacement of 3.5 liters.
- P14-18 Four-stroke inline seven-cylinder with a displacement of 3.5 liters.
- P14-19 Two-stroke inline eight-cylinder with a displacement of 4 liters.
- P14-20 Four-stroke inline eight-cylinder with a displacement of 4 liters.
- P14-21 Two-stroke vee ten-cylinder with a displacement of 5 liters.
- P14-22 Four-stroke vee ten-cylinder with a displacement of 5 liters.
- P14-23 Four-stroke W-6 comprised of three banks of two with a displacement of 5 liters.*
- P14-24 Four-stroke W-9 comprised of three banks of three with a displacement of 5 liters.*
- P14-25 Four-stroke W-12 comprised of three banks of four with a displacement of 5 liters.*
- P14-26 Four-stroke W-12 comprised of four banks of three with a displacement of 5 liters.*
- P14-27 Design a family of vee engines, all with same pistons, connecting rods, and strokes. Crankshafts can each be different. Four configurations are needed: vee-four, vee-six, vee-eight, and vee-ten with the same single-cylinder displacement of 0.5 liters. Optimize the single-cylinder configuration from which the multicylinder engines will be constructed for bore/stroke ratio and conrod/crank ratio. Then assemble this cylinder design into the above configurations. Find the best compromise of vee angle to provide a good mix of balance and even firing in all engines.
- P14-28 Repeat Project P14-27 for a family of three-bank W engines: W-3, W-6, W-9, and W-12. The interbank angles must be the same for all models. See the built-in example W-12 engine in program LINKAGES for more information on this unusual W configuration.
- P14-29 In recent years some automobile manufacturers have made unusual vee configurations such as the VW-Audi VR15 which is a 15° vee-six. Obtain detailed information on this engine design and then analyze it with program LINKAGES. Write a report that explains why the manufacturer chose this unusual arrangement and justify your conclusions with sound engineering analysis.
- P14-30 Design an inline six- and an inline five-cylinder engine of the same displacement, say 2.5 liters. Analyze their dynamics with program LINKAGES. Write an engineering report to explain why such manufacturers as Audi, Volvo, and Acura have chosen a five-cylinder inline over a six-cylinder of comparable torque and power output.
- P14-31 Ferrari has produced vee-twelve engines in both 60° vee and horizontally opposed configurations. Design 3-liter versions of each and compare their dynamics. Write a report that explains why the manufacturer chose these arrangements and justify your conclusions with sound engineering analysis.
- P14-32 Design and compare a 3-liter 90° vee-six, 60° vee-six, inline six, and 180° opposed six, examples of which are all in volume production. Explain their advantages and disadvantages and justify your conclusions with sound engineering analysis.

* A W engine has three banks of four cylinders or four banks of three cylinders on a common crankshaft. The VW-Audi W-12, as used in some Bentley models, is two VR15, 15° vee sixes grafted together on a common crankshaft.